
Coordination and Conformity in Multi-Option LLM Systems

MASTER'S THESIS

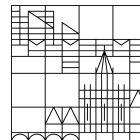
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Lorenz Rückert

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Department of Politics and Public Administration

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1. Reviewer: Dr. Giordano De Marzo
2. Reviewer: Prof. Dr. Susumu Shikano

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Abstract

This thesis studies how coordination among large language model (LLM) agents changes when more than two opinions are available. Opinion dynamics simulations of six open-weight models, with 25 to 800 agents and 3 to 50 opinions, are combined with controlled response measurements from about 254,000 queries, using neutral labels and a memoryless interaction protocol. The mean-field Potts softmax extends the binary Curie-Weiss response and is used to relate individual choices to collective outcomes.

Early coordination is rapid in most models. Strong leads form within a few time steps within the observed timeframe and across the tested numbers of opinions. Many opinions disappear early even in runs that later stall. Additional opinions do not impose a uniform coordination cost. Both Gemma models reach consensus in every tested condition up to 800 agents, while the Qwen models become slow or stop finishing beyond a model-specific number of opinions that decreases with population size; the Llama populations retain disagreement at every number of opinions. Among conditions with reliable completion, consensus time increases more consistently with population size than with the number of opinions.

The Potts softmax describes the Gemma and Qwen choices within a condition, but its fitted conformity strength grows with the number of opinions, and at many opinions the conformity estimated from individual states declines as the leader gains support, whereas the binary response of the same model is compatible with a single constant. The Llama responses depart more strongly from the rule. The theoretical benchmarks do not reliably separate completed from unfinished runs, but the measured responses retain information about coordinating behaviour beyond the fitted parameter. Additionally, a binary measurement made with the labels that the two surviving opinions carry accounts for the prolonged disagreement after extinction has removed the others. The binary framework can thus describe the late stage of multi-option coordination.

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List of Abbreviations

AI	A rtificial I ntelligence
API	A pplication P rogramming I nterface
AWQ	A ctivation-aware W eight Q uantization
GPU	G raphics P rocessing U nit
LLM	L arge L anguage M odel
MMLU-Pro	E xtended version of the M assive M ultitask L anguage U nderstanding benchmark
RMSE	R oot M ean S quared E rror

List of Symbols

N	group size (number of agents)
q	number of opinions
m	binary collective opinion
$C(t)$	coordination level (Potts order parameter); equals $ m $ for $q = 2$
$m(t)$	vector of opinion shares
$m_\alpha(t)$	population share supporting option α
$s(t)$	leader share (share of the current plurality)
$P(m)$	binary adoption probability
$P(\alpha \mid m)$	probability of adopting option α in state m
$P_{\text{lead}}(s)$	probability of choosing the leading option
β	conformity strength (majority force)
$F_{\text{in}}(s)$	expected inflow into the leading option
$F_{\text{out}}(s)$	expected outflow from the leading option
$\nu(s)$	reduced drift of the leader share
s^*	resting point of the reduced drift
$\beta_{\text{bal}}(q)$	balance-instability threshold, $\beta_{\text{bal}}(q) = q/2$
$\beta_c(q)$	critical value of the mean-field Potts model
$\beta_{\text{eff}}(s)$	effective conformity at a single state
$\beta_t(N, q)$	fluctuation threshold for completing consensus
N_c	critical group size of the binary model
r	relative leader position of a controlled state
$K(t)$	number of active opinions at time t
$T(N)$	consensus time as a function of group size
t	time steps t in sweeps

Chapter 1

Introduction

1.1 Motivation

Large language models (LLMs) are increasingly used as agents: systems that pursue tasks over many steps, call tools, and interact with other agents rather than only with a person (Guo et al., 2024). Software teams simulated entirely by language models write code together (Qian et al., 2024), frameworks organize conversations among multiple model instances (Q. Wu et al., 2023), populations of generative agents are used to simulate small societies (Park et al., 2023), and agent-based applications now reach specialized domains such as medicine (Y. Lin et al., 2025). Moltbook provides another example of agents in a group setting. It is organized as a social network in which artificial intelligence (AI) agents interact while humans mainly observe (Jiang et al., 2026). These systems shift the focus of the analysis. Alongside the output of a single model, attention turns to what emerges from repeated interactions among models in a group.

For populations of LLMs, coordination is an important capability that can have positive as well as negative consequences. Agent systems need to agree on shared choices to cooperate, but agreement alone does not ensure a good decision. A population that converges too readily may settle on an arbitrary or undesirable choice. Experiments with pricing algorithms powered by AI for example, show how repeated interactions can lead agents to sustain prices above competitive levels (Calvano et al., 2020). Such outcomes illustrate that coordination among agents can have harmful consequences for others. Similar concerns have been studied in human groups through concepts such as groupthink or informational cascades, where social influence can restrain dissent or cause an early majority to become self-reinforcing

(Banerjee, 1992; Bikhchandani et al., 1992; Janis, 1982). Both sides make it necessary to understand when and how populations of language-model agents coordinate.

An efficient way to study such questions comes from modeling traditions that explain group outcomes through simple rules followed by many interacting individuals. Examples include threshold models of collective behavior (Granovetter, 1978) and statistical-physics models of social processes (Castellano et al., 2009). That humans tend to follow majorities has long been documented in social psychology (Asch, 1956). The models add a macroscopic perspective by explaining what such individual tendencies can produce at the level of a whole group.

This approach has recently been applied directly to LLMs. De Marzo, Castellano, et al. (2026) placed groups of LLM agents in a repeated binary-choice setting, where each agent sees the current opinions of all others and picks one of two options. For most tested models, the response to a majority could be summarized by one parameter, the majority force, using the response rule of the mean-field Ising, or Curie-Weiss, model. Their work suggests that some aspects of behavior in LLM populations can be described through a mathematical rule, even when the internal processes behind an individual model’s choices remain unclear. Follow-up work introduced a second parameter, an intrinsic bias h toward one option, and showed that the competition between bias and conformity can produce collective choices that differ from the models’ intrinsic preferences (De Marzo, Bellina, et al., 2026). This use of behavioral and mathematical tools to study artificial populations follows the perspective of collective machine behavior (Rahwan et al., 2019), as applied to LLM populations by De Marzo, Castellano, et al. (2026). The present thesis adopts this perspective and investigates coordination when agents choose among multiple opinions.

1.2 Related Work

1.2.1 Coordination and social influence

Several related concepts are relevant to this thesis. Coordination concerns how individuals arrive at compatible choices when several alternatives are available, while a convention is an established pattern of choice maintained by expectations about what others will do (Lewis, 2008; Schelling, 1980). Following De Marzo, Castellano,

et al. (2026), this thesis distinguishes between partial and full coordination. Full coordination, also called consensus, is reached when every agent supports the same opinion. Partial coordination means that support is concentrated on a leading opinion, but not all agents agree. Coordination is also used more loosely to describe the process and degree of agreement. Conformity describes the individual tendency to follow others. A tendency to follow a majority does not by itself tell us whether the whole population will eventually agree.

Research on human groups provides evidence for this individual tendency and its consequences. Asch’s experiments showed that people sometimes adopt an incorrect majority answer even in a simple judgment task (Asch, 1956). Later research found that the strength of the effect varies across settings and cultures, and distinguished between following others as a source of information and following them to gain social acceptance (Bond & Smith, 1996; Cialdini & Goldstein, 2004). Social influence can also reduce diversity in group estimates and amplify early signals of approval (Lorenz et al., 2011; Muchnik et al., 2013). At the collective level, experiments show that people can develop shared conventions through local interactions, and that committed minorities can overturn an established convention (Centola et al., 2018; Centola & Baronchelli, 2015). These studies connect individual responses to group outcomes, which is also the central question in the present thesis. They do not imply that language models follow others for the same psychological reasons.

Conformity has also been measured directly in LLMs. Models presented with a question and a majority answer sometimes adopt that answer, with the response depending on the majority size and on properties of the model and task (Weng et al., 2025; Zhu et al., 2025). Adaptations of Asch’s visual tasks report related effects in multimodal models (Bellina et al., 2026). Studies of multi-agent debate also distinguish agreement from accuracy. Additional discussion does not consistently improve answers, and convergence can result from selective persuasion and not from a general tendency to agree (Cau et al., 2025; Kaesberg et al., 2025). In simulations of discussion about meaningful topics, both evidence and induced biases can shape the position on which agents converge (Chuang et al., 2024). These settings involve options with substantive content. The main experiments of this thesis instead use neutral, arbitrary labels without arguments for or against them which allows the response to the distribution of support to be studied directly.

1.2.2 Models of collective behavior

Mathematical models provide one way to connect such responses to the behavior of a population. Threshold models describe individuals who act once enough others have acted (Granovetter, 1978). Models from statistical physics describe agents whose states change according to simple update rules which makes it possible for researchers to study when agreement emerges, when opinions coexist, and how long these processes take (Baronchelli, 2018; Castellano et al., 2009). The Curie-Weiss model describes two possible states. Its multi-state counterpart, the mean-field Potts model, allows more than two (F. Y. Wu, 1982, 1984). Under the corresponding Glauber update rule, each individual selects a state probabilistically according to the current configuration (Cuff et al., 2012; Glauber, 1963). In the opinion setting, this becomes a rule that converts the support for each option into an adoption probability.

The number of opinions changes the theoretical behavior of the Potts model. With two options, the preferred equilibrium moves gradually away from equal support as conformity increases beyond a critical value. With three or more, it shifts discontinuously to an ordered state, in which one opinion has more support than the others. Such a state does not represent consensus in every case. A population can remain close to balance even when an ordered state is more favorable in equilibrium (Costeniuc et al., 2005; Cuff et al., 2012; Ellis & Wang, 1990). The distinction between a preferred equilibrium and the time needed to reach it is relevant when comparing theory with finite simulations. Chapter 2 develops these distinctions for the notation used in this thesis.

Other update rules lead to different expectations. In the multi-state voter model, where an agent copies another agent's opinion, the mean-field consensus time increases only moderately with the number of initially represented opinions and approaches a limiting value (Starnini et al., 2012). The naming game uses a different mechanism. Pairs of agents exchange names, retain information from their interactions, and develop a shared convention through successful exchanges (Baronchelli et al., 2006; Baronchelli, 2018). According to the findings, additional opinions do not impose the same difficulty in every model. The effect depends on how individuals respond and on the information available to them.

1.2.3 Coordination in LLM populations

Empirical work on LLM populations examines both open-ended interaction and controlled coordination tasks. Analyses of Moltbook, for example, describe patterns of activity and popularity across an agent social network (De Marzo & Garcia, 2026; Jiang et al., 2026). Controlled experiments address a more specific question, namely, how repeated choices by individual agents produce agreement across a population. The binary study of De Marzo, Castellano, et al. (2026) provides the main starting point for this thesis. An agent is shown the current opinions of all other agents and chooses between two options, without retaining a history of previous interactions. Repeating queries at controlled opinion distributions, as in the bias study (De Marzo, Bellina, et al., 2026), allows the probability of adopting either option to be estimated directly.

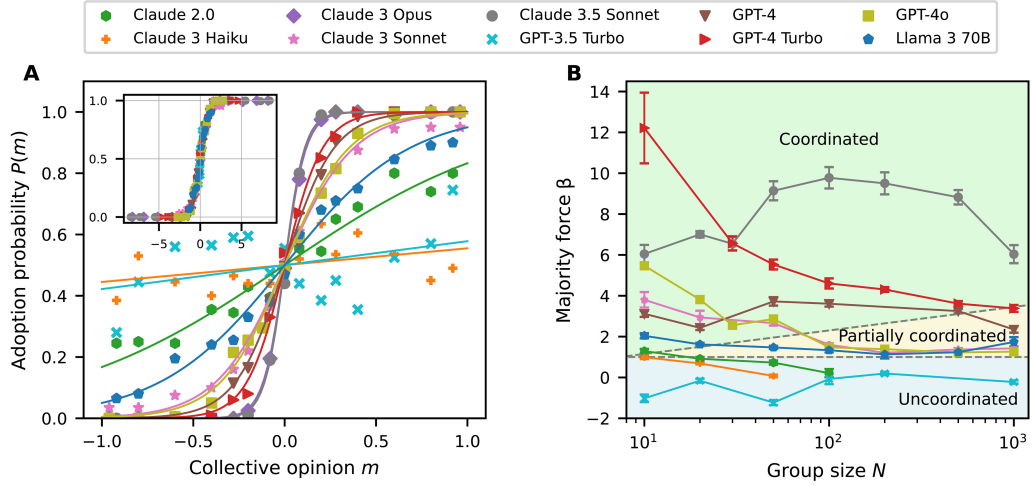


FIGURE 1.1: Individual response at $N = 50$ (A) and phase diagram (B) of the binary study. (A) Adoption probability $P(m)$ against the collective opinion m for ten LLMs, with fits of $P(m) = \frac{1}{2}[\tanh(\beta m) + 1]$. (B) Fitted majority force β against group size N ; the horizontal dashed line marks $\beta_c = 1$ and the rising dashed line the crossover point $\beta_t(N)$ above which fluctuations lead to coordination.

Note. From “AI Agents Can Coordinate via Majority-Following Beyond Human Scale” (Figure 3), by G. De Marzo, C. Castellano, and D. Garcia, 2026, *Science Advances*, 12(33), Article eaea6091 (<https://doi.org/10.1126/sciadv.aea6091>). Copyright 2026 by the authors. Reproduced under CC BY-NC 4.0.

For most models in that study, the measured response is approximately described

by the hyperbolic-tangent rule of the Curie-Weiss model (Figure 1.1 A). The fitted conformity strength can then be compared with theoretical benchmarks for the formation of a majority and the completion of consensus. The authors report differences across models and group sizes, including an association between model capability and the largest populations able to reach full coordination (De Marzo, Castellano, et al., 2026).

The bias study applies the same approach to binary choices with meaningful content. It adds an intrinsic preference to the response rule and studies how this preference competes with majority influence. The resulting populations can continue to support an option that runs against the models' intrinsic preference (De Marzo, Bellina, et al., 2026). Together, the two studies lay out why conformity and intrinsic preference need to be distinguished. They also provide the estimation logic used here which is measuring individual adoption probabilities under controlled conditions, fit a simple response rule, and compare its implications with opinion dynamics simulations.

Multi-option coordination has also been studied through naming games. Ashery et al. (2025) show that LLM populations can develop shared conventions through pairwise interactions and limited memory. In their setup, two randomly paired agents each pick a name from a shared pool, an interaction counts as successful when both pick the same name, and each agent remembers only its last few interactions. Figure 1.2 shows the success rate rising to one within about fifteen population rounds for three of the four models. Their experiments also find collective preferences that are not deducible from isolated choices. Follow-up work extracts the agents' choice policies and uses a mean-field description to analyze collective states and group-size effects but the authors restrict themselves to two conventions (Flint et al., 2026). Another preprint studies how learning from interactions contributes to the tendency of such populations to adopt one label (Tanaka, 2026).

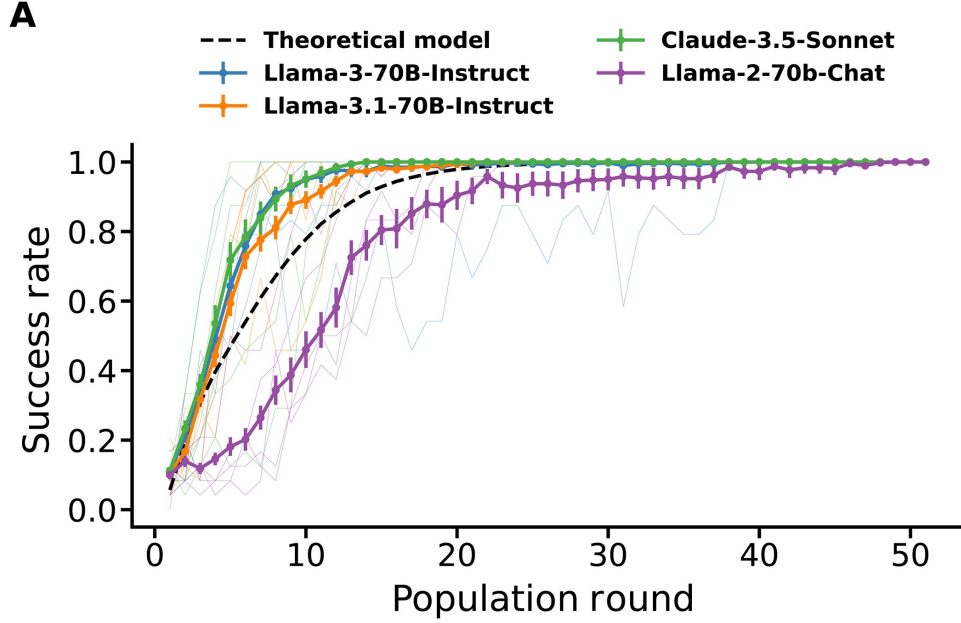


FIGURE 1.2: Emergence of a shared convention in the naming game. Success rate, the probability that a pairwise interaction ends with both agents producing the same name, by population round for $N = 24$ agents and a pool of $W = 10$ names. Thick lines average 40 runs per model, thin lines show individual runs, and the dashed line is the minimal naming-game model.

Note. Adapted from “Emergent Social Conventions and Collective Bias in LLM Populations” (Figure 1, panel A), by A. F. Ashery, L. M. Aiello, and A. Baronchelli, 2025, *Science Advances*, 11(20), Article eadu9368 (<https://doi.org/10.1126/sciadv.adu9368>). Copyright 2025 by the authors. Reproduced under CC BY-NC 4.0.

1.2.4 Preferences and experimental design

The way opinions are presented matters for both protocols. LLMs can prefer particular option labels or positions in a multiple-choice prompt (Pezeshkpour & Hruschka, 2024; Zheng et al., 2024). When the choices have substantive content, measured preferences also depend on the model, the language, the available answer options and the wording of the question (Buyl et al., 2026; Röttger et al., 2024; Santurkar et al., 2023). Training with human feedback provides one route through which model responses are shaped toward particular preferences (Ouyang et al., 2022). Neutral opinion labels reduce the role of substantive disagreements, but they do not ensure that models are indifferent to the displayed tokens.

Randomizing which label is assigned to a specific opinion prevents a preference for a particular label from systematically favoring the same opinion. It does not establish that label preferences disappear from the measured response. This matters when comparing different numbers of opinions, because the label pool also changes. More generally, reviews of generative social simulations point out the sensitivity of results to prompts and interaction protocols and the need for validation (Larooij & Törnberg, 2025). Similarities between the behavior of LLMs and people should also not be taken as evidence that the underlying mechanisms are the same (Z. Lin, 2025). The present thesis studies the models themselves. Its conclusions concern their responses under the specified experimental conditions, and the transfer from controlled measurements to opinion dynamics is examined empirically.

1.3 Research Gap

Many group decisions involve more than two options, such as choosing a name, a rule of behavior, a candidate or a method for solving a problem. In a binary setting, knowing the support for one opinion also determines the support for the other. With several opinions, the same leader share can occur alongside different compositions. With almost equal support, the leading opinion may also represent only a small fraction of the population. Extending the binary setting consequently changes both the information an agent receives and the range of distributions through which the population can move.

The binary study provides a response framework, an individually measured adoption rule under full information about the group together with thresholds that link it to collective outcomes, but only for two opinions. Multi-option coordination has been studied in naming games, where agents interact in pairs and remember past exchanges; its mean-field treatment based on measured policies is also restricted to two conventions (Flint et al., 2026). The combination of the two, a measured response to more than two opinions under full information and its connection to population dynamics through the Potts extension of the binary framework, has not been examined as far as the literature reviewed here shows. The Potts softmax provides a possible rule for this response. It assigns adoption probabilities from opinion shares using a single conformity strength, and its thresholds give benchmarks for the emergence and completion of coordination. Whether that rule describes

LLM choices, and whether its fitted parameter remains meaningful across different distributions and numbers of opinions, are empirical questions.

This leaves three connected issues. First, it is unclear whether additional opinions make coordination slower or less likely. Second, it is unclear whether a model's majority-following response can be summarized by the same conformity strength as the number and distribution of opinions change. Third, even if the Potts softmax does not capture the response fully, the measured adoption probabilities may still help explain why an opinion grows or why a population fails to reach consensus. Answering this last question requires comparing controlled response measurements with the behavior of interacting populations.

The thesis addresses these issues through two types of experiments using six open-weight models from three model families. Opinion dynamics simulations record how initially balanced populations change across numbers of opinions and group sizes. Controlled experiments repeatedly query the same controlled distributions to estimate individual adoption probabilities. Most controlled states vary one leading opinion's share while dividing the remaining support as evenly as possible. This provides a consistent basis for comparison, while limiting which distributions are measured directly. Comparisons with the simulations and supplementary measurements examine how far the resulting description applies beyond those states. The Potts model supplies a theoretical approximation whose usefulness and limitations are assessed through these observations.

1.4 Research Questions and Hypotheses

The three research questions follow from these gaps. Each research question is accompanied by a hypothesis motivated by the theoretical framework of Chapter 2 and the binary study. The hypotheses guide the analysis and are assessed through the descriptive comparisons and model checks of Chapter 4. No formal significance tests are used to evaluate them. Statements of support describe agreement between the observations and the expectations within the tested conditions. Partial support indicates that this agreement varies across models or conditions. Where an expectation is not supported, the analysis distinguishes contrary observations from insufficient evidence.

RQ1: Does coordination among LLM agent populations become more difficult as the number of available opinions increases?

The Potts model motivates this question because the conformity strength required to make an ordered equilibrium preferable increases with the number of opinions. If conformity does not increase accordingly, additional opinions may make agreement more difficult.

H1: Additional opinions make consensus more difficult. For a given model and group size, increasing the number of opinions is expected to make consensus slower or less likely within the observation horizon. The analysis compares completion and consensus times across q at fixed N and uses the fitted conformity strengths to examine the theoretical explanation.

RQ2: How does the strength of majority-following behavior change with the number of available opinions when responses are measured under controlled conditions?

Changes in conformity as opinions are added may help explain the coordination outcomes under RQ1. The controlled measurements examine both changes across the number of opinions and changes along each response curve.

H2: One β describes each model’s conformity. For a given model and displayed group size, the Potts softmax is expected to describe responses across opinion shares with one conformity strength β , and this value is expected to remain approximately constant across the number of opinions. These are two distinct checks. Whether one parameter describes a whole measured curve, and whether its fitted value transfers across numbers of opinions. Constancy across group sizes is not assumed, since the binary study already reports a dependence on group size (De Marzo, Castellano, et al., 2026).

RQ3: Does the Potts softmax adequately describe multi-option conformity in LLM systems, and can the empirically measured response function predict collective behavior?

A fitted curve can describe controlled choices without accounting for the states visited in an opinion dynamics simulation. Conversely, a measured response may still explain collective behavior even when the fitted curve is imperfect. Thus, RQ3 examines the connection between the individual response and the dynamics it produces.

H3: Individual responses predict collective dynamics. The measured adoption probabilities are expected to indicate whether a leading opinion gains or loses support and whether the population continues toward consensus or remains partially coordinated. This is examined through the drift construction developed in Chapter 2, which compares the probability of adopting an opinion with its current share. The comparison also tests how far measurements from the controlled states transfer to the less regular distributions that arise during simulations.

1.5 Thesis Structure

Chapter 2 develops the conceptual framework. It introduces the binary model and its multi-option extension, then connects the individual response rule to collective dynamics through the flow of supporters, the stability of the balanced state and possible states of partial coordination. It provides the theoretical basis for the hypotheses introduced above.

Chapter 3 describes the opinion dynamics simulations, the controlled response measurements and the binary consistency checks. Chapter 4 first describes the observed coordination behavior across models, numbers of opinions and group sizes, and then examines how far the measured responses account for it. Chapter 5 discusses the findings in relation to the research questions and existing literature, and considers the limitations of the study. Chapter 6 concludes the thesis.

Chapter 2

Theoretical Framework

2.1 The Binary Conformity Model

This thesis builds on the binary framework of De Marzo, Castellano, et al. (2026). Their setup is a population of N agents, each holding one of two opinions. At each update, one agent is selected at random, shown the opinions of all other agents and asked which opinion it supports. Its state is then updated accordingly. The agents have no memory of previous interactions and are not told their own current opinion. One time unit, or sweep, corresponds to N individual updates. The state of the population is summarized by the collective opinion

$$m = \frac{N_+ - N_-}{N},$$

where N_+ and N_- count the supporters of the two opinions. The coordination level is $C = |m|$, with $C = 1$ representing full coordination (consensus) and $C = 0$ an evenly balanced population.

For most tested models, the probability of adopting the first opinion approximately follows

$$P(m) = \frac{1}{2} [\tanh(\beta m) + 1],$$

where β is the conformity strength, called the majority force in the binary study. For $\beta = 0$, the agent chooses at random; as β increases, it follows the current majority more deterministically. Fitted values differ across models and group sizes. At fixed N , more capable models generally exhibit a larger conformity strength,

while for most models in the binary study β tends to decrease as N increases (De Marzo, Castellano, et al., 2026). The responses are measured with shuffled labels and ordering.

The response has the same form as the mean-field Glauber update probability of the Curie–Weiss model (Castellano et al., 2009; Glauber, 1963). In the mean-field model, the stationary collective opinion satisfies

$$m^* = \tanh(\beta m^*).$$

For $\beta \leq 1$, the only solution is $m^* = 0$. Above $\beta_c = 1$, this balanced solution becomes unstable and two stable ordered solutions with opposite signs emerge. Their magnitude increases smoothly as β grows. Order consequently develops gradually and is not appearing with a sudden large imbalance, which is why this is called a continuous phase transition (Castellano et al., 2009).

An ordered state does not mean consensus at $m = \pm 1$. In a finite population, the collective opinion fluctuates around m^* . The binary framework distinguishes three regimes: near balance for $\beta \leq 1$, partial coordination for intermediate conformity strength, and a regime in which ordinary fluctuations can complete consensus. The crossover to this last regime is approximately

$$\beta_t(N) \approx \frac{1}{2} \ln N.$$

Below this crossover, an ordered population can remain below consensus for a long time, with completion becoming increasingly difficult as N grows. Above it, the ordered state is sufficiently close to consensus for ordinary fluctuations to complete the process (De Marzo, Castellano, et al., 2026).

If the measured conformity strength crosses this rising threshold, a practical critical group size N_c is defined by

$$\beta(N_c) = \beta_t(N_c).$$

Beyond the crossing, full coordination becomes difficult to reach on ordinary timescales.

2.2 Multi-Option Generalization

With q opinions, the state of the population is a vector of opinion shares,

$$m = (m_1, \dots, m_q), \quad m_\alpha \geq 0, \quad \sum_{\alpha=1}^q m_\alpha = 1.$$

At balance, every opinion has share $1/q$; at consensus, one share equals one. The share of the currently leading opinion is $s(t) = \max_\alpha m_\alpha(t)$. To compare coordination across different numbers of opinions, this share is rescaled as

$$C(t) = \frac{q s(t) - 1}{q - 1}.$$

It equals zero at balance and one at consensus. For $q = 2$, it reduces to the binary coordination level $|m|$. This is the scalar Potts order parameter used in the figures (F. Y. Wu, 1982).

The multi-option counterpart of the Curie–Weiss response is the mean-field Potts softmax (Cuff et al., 2012; F. Y. Wu, 1982, 1984),

$$P(\alpha \mid m) = \frac{\exp(2\beta m_\alpha)}{\sum_{\gamma=1}^q \exp(2\beta m_\gamma)}.$$

An updating agent weighs every option by the exponential of its current share, scaled by 2β . The factor two keeps β on the scale of the binary conformity strength. The rule is the mean-field heat-bath response connected to Potts Glauber dynamics. Here it is a candidate description of LLM choices, whose accuracy must be checked against the measurements.

The rule treats all options symmetrically, so an option is favored only when it has more support. The relative odds between any two options depend only on the difference between their shares. Thus, a five-percentage-point advantage produces the same relative odds between those two options regardless of how many other options are present. The other options still affect their individual adoption probabilities through the denominator. One constant β is assumed to convert share differences into relative odds across all these states.

For $q = 2$, writing $m_2 = 1 - m_1$ and $m = m_1 - m_2$ gives

$$\begin{aligned} P(1 \mid m) &= \frac{e^{2\beta m_1}}{e^{2\beta m_1} + e^{2\beta m_2}} \\ &= \frac{1}{1 + e^{-2\beta(m_1 - m_2)}} \\ &= \frac{1}{2} [\tanh(\beta m) + 1]. \end{aligned}$$

The binary response is recovered with the same β , so no conversion is needed when comparing values with De Marzo, Castellano, et al. (2026) and De Marzo, Bellina, et al. (2026).

An ordered state is a configuration in which one opinion has more support than the others. It doesn't need to represent consensus. The critical value of the mean-field Potts model identifies when an ordered state becomes preferred over balance in equilibrium. In the normalization used here, it is

$$\beta_c(q) = \frac{q-1}{q-2} \ln(q-1), \quad q \geq 3,$$

with limiting value $\beta_c(2) = 1$ (Costeniuc et al., 2005; Cuff et al., 2012; Ellis & Wang, 1990). For $q \geq 3$, the transition is first-order. The preferred leader share changes discontinuously. At $\beta_c(q)$, the balanced state and ordered states are equally favored. The former has leader share $1/q$, while the latter already has leader share $(q-1)/q$. The preferred imbalance consequently appears with a finite size instead of growing gradually from balance.

The critical value grows only logarithmically with q : $\beta_c(3) \approx 1.39$, $\beta_c(10) \approx 2.47$ and $\beta_c(50) \approx 3.97$. Its increase provides the equilibrium motivation for H1 when conformity is held fixed. Below it, balance is preferred; above it, an ordered state is preferred. This does not determine how quickly either state is reached. In the first-order case, ordered states can remain locally stable below the critical value, and balance can remain locally stable above it (Cuff et al., 2012). The next section distinguishes this equilibrium preference from the emergence of a leader and the completion of consensus.

2.3 From Individual Responses to Collective Dynamics

The response rule describes an individual choice. To connect it to population behavior, this section follows the drift argument used by De Marzo, Castellano, et al. (2026). The calculation first identifies the direction in which support changes, then applies the Potts response near balance and near consensus.

2.3.1 The Drift of the Leading Opinion

A leader share does not specify the complete population state, the same s can occur alongside one strong competitor or several weaker ones. In general, the adoption probability depends on the full share vector m . The notation $P_{\text{lead}}(s)$ refers here to the reduced response measured along the controlled states of Chapter 3, where the remaining support is divided as evenly as possible. Each response curve is specific to a model, amount of opinions and a displayed group size.

To derive the flow, consider one opinion L that leads in the current state, with share $s = m_L$. The calculation tracks support for this opinion. While it remains the leader, its share is also the largest share; at ties or changes of leader, the drift of the maximum doesn't need to equal the drift of this particular opinion.

The mean-field description assumes that choices depend on the displayed distribution, not on the updating agent's history or its own opinion. It also neglects the one-agent difference between the distributions shown when different agents are selected.¹ The same adoption probability can then be used for agents joining and leaving the leading opinion. The expected inflow and outflow per update are

$$\begin{aligned} F_{\text{in}}(s) &= (1 - s)P_{\text{lead}}(s), \\ F_{\text{out}}(s) &= s[1 - P_{\text{lead}}(s)]. \end{aligned}$$

¹If an agent holding opinion γ is selected, the displayed share vector is $(Nm - e_\gamma)/(N - 1)$, where e_γ removes that agent's supporter count. The exact finite-population drift averages responses over these possible displays. The mean-field approximation neglects their differences; its accuracy depends on how strongly the response changes between neighboring count distributions.

The first expression multiplies the probability of selecting a non-supporter by the probability that it adopts the leader. The second multiplies the probability of selecting a supporter by the probability that it chooses another opinion. Their difference is the reduced drift,

$$\nu(s) = F_{\text{in}}(s) - F_{\text{out}}(s) = P_{\text{lead}}(s) - s.$$

Per individual update, the expected change in share is $\nu(s)/N$; in time measured in sweeps, the corresponding mean-field equation is $ds/dt = \nu(s)$. Comparable to the multi-option version of the right- and left-transition probabilities $R(m)$ and $L(m)$ used in the binary framework.²

Where $\nu(s) > 0$, support for the opinion grows on average; where $\nu(s) < 0$, it shrinks. At a crossing

$$P_{\text{lead}}(s^*) = s^*,$$

the two flows balance. If the drift is positive below the crossing and negative above it, support is pushed back toward s^* from both sides. This gives a stable resting point of the reduced description and a candidate partially coordinated state. If the directions are reversed, the crossing is unstable.

Zero drift does not mean that agents stop changing their opinions. At an interior resting point, they continue to switch in both directions, and finite-population fluctuations remain. The simulations instead stop at consensus, treating full coordination as the endpoint of a run. A crossing also need not be an equilibrium of the complete share vector: its predictive value depends on whether the measured response applies to the distributions visited during the simulation.

The drift can be calculated directly from measured adoption probabilities and does not require a fitted Potts curve. Its usefulness goes beyond the controlled slice but still requires an empirical check. For the symmetric response at exact balance, each designated opinion has adoption probability $1/q$, so

²In the binary notation, $R(m) = \frac{1-m}{2}P(m)$ and $L(m) = \frac{1+m}{2}[1 - P(m)]$. For a tracked opinion with share $s = (1+m)/2$, these equal the inflow and outflow above, and $dm/dt = 2 ds/dt$ (De Marzo, Castellano, et al., 2026).

$$P_{\text{lead}}(1/q) = \frac{1}{q}, \quad \nu(1/q) = 0.$$

Randomized label assignments target the same symmetry in the controlled measurements, although integer counts may permit only approximate balance. Zero drift for each designated opinion does not prevent fluctuations from creating a leader. To determine whether a small lead tends to grow or disappear, the response must be examined slightly away from balance.

2.3.2 Stability of the Balanced State

The mean-field drift of the Potts Glauber dynamics can be expanded close to balance (Cuff et al., 2012). Write a small deviation for each opinion as

$$m_\alpha = \frac{1}{q} + \varepsilon_\alpha, \quad \sum_{\alpha=1}^q \varepsilon_\alpha = 0.$$

The second condition ensures that the shares still sum to one. If one opinion gains support, the others must collectively lose the same amount. Substituting these shares into the softmax gives

$$P(\alpha \mid m) = \frac{e^{2\beta(1/q + \varepsilon_\alpha)}}{\sum_{\gamma=1}^q e^{2\beta(1/q + \varepsilon_\gamma)}}.$$

Close to balance, $2\beta\varepsilon_\alpha$ is small, so the first-order Taylor approximation $e^x \approx 1 + x$ can be used. The common factor $e^{2\beta/q}$ cancels from the numerator and denominator. The first-order changes in the denominator also cancel because the deviations sum to zero. Thus,

$$P(\alpha \mid m) \approx \frac{1 + 2\beta\varepsilon_\alpha}{q} = \frac{1}{q} + \frac{2\beta}{q}\varepsilon_\alpha.$$

The current deviation from balance is ε_α , whereas the deviation in the adoption probability is $(2\beta/q)\varepsilon_\alpha$. Comparing the two gives

$$P(\alpha \mid m) - m_\alpha \approx \left(\frac{2\beta}{q} - 1 \right) \varepsilon_\alpha.$$

With time measured in sweeps, this becomes

$$\frac{d\varepsilon_\alpha}{dt} \approx \left(\frac{2\beta}{q} - 1 \right) \varepsilon_\alpha.$$

When $\beta < q/2$, small deviations shrink and the population is pushed back toward balance. When $\beta > q/2$, they grow and one opinion can begin to emerge as the leader. The boundary is the balance-instability threshold,

$$\beta_{\text{bal}}(q) = \frac{q}{2}.$$

At equality, the first-order drift vanishes. This calculation uses the complete share vector and does not require the other opinions to have equal shares. It is still a local result. It concerns very small deviations from balance, not the entire path to consensus.

For $q = 2$, $\beta_{\text{bal}} = \beta_c = 1$, recovering the binary critical value. For $q \geq 3$, the two benchmarks separate. When $\beta_c(q) < \beta < \beta_{\text{bal}}(q)$, an ordered state is preferred in equilibrium, but small deviations from balance still shrink. Balance is then metastable: the population can remain near a locally stable state even though another state is globally more favorable (Cuff et al., 2012). Escaping requires a sufficiently large fluctuation, whose likelihood depends on the group size and the position of β within this interval.

The threshold's direct use is limited by this for the experiments. Population shares change in steps of $1/N$, which may already exceed the small deviations assumed here when each opinion has few supporters. The controlled measurements also use finite leads and do not directly measure the slope at exact balance. Because of this, a pooled fit below β_{bal} does not mean that the simulated population should remain near balance.

2.3.3 The Fluctuation Threshold

Once a population is ordered, one question remains: can it complete the last steps toward consensus? The binary fluctuation threshold $\beta_t(N) \approx \frac{1}{2} \ln N$ addresses this stage. A counting argument gives a corresponding approximation for the fixed- q Potts model.

Near consensus, the leading opinion has share close to one and the other shares are close to zero. Due to this, the leader's softmax weight is approximately $e^{2\beta}$, while each of the $q - 1$ other opinions has weight approximately $e^0 = 1$. Adding the weights of all non-leading opinions gives their total adoption probability,

$$P_{\text{dissent}} = \frac{\sum_{\gamma \neq L} e^{2\beta m_\gamma}}{\sum_{\alpha=1}^q e^{2\beta m_\alpha}} \approx \frac{q-1}{e^{2\beta} + q-1} \approx (q-1)e^{-2\beta}.$$

The last approximation neglects the $q - 1$ term in the denominator and requires $e^{2\beta} \gg q - 1$. During one sweep, the expected number of non-leader choices near this state is approximately NP_{dissent} . If this is much larger than one, dissent is continually regenerated. If it is of order one or smaller, full coordination becomes easier to reach through fluctuations. Setting it approximately equal to one gives

$$N(q-1)e^{-2\beta_t} \approx 1 \quad \implies \quad \beta_t(N, q) \approx \frac{1}{2} \ln [(q-1)N].$$

For $q = 2$, this recovers the binary scaling of De Marzo, Castellano, et al. (2026). The calculation is a leading-order completion benchmark, not an exact transition or a guarantee of consensus. It assumes a near-consensus state and a fixed set of q available opinions; in the theoretical Potts model, even opinions with zero supporters retain a non-zero adoption probability.

The prompt protocol differs at this point. Unsupported opinions disappear from the displayed list, so the number of active opinions can fall to $K(t) < q$. If absent opinions are effectively unavailable and the remaining choices follow the same softmax form, the argument can be applied with $K(t)$ in place of q . When two opinions remain, the binary framework provides a natural completion benchmark.

The three theoretical scales answer different questions:

TABLE 2.1: The three theoretical scales and the questions they address.

Quantity	Question
$\beta_c(q)$	Is an ordered state preferred over balance in equilibrium?

Quantity	Question
$\beta_{\text{bal}}(q) = q/2$	Does a small deviation from balance grow or disappear?
$\beta_t(N, q)$	Near an ordered state, is dissent sufficiently weak for fluctuations to complete consensus?

There is no universal ordering between β_t and β_{bal} . For example, at $q = 10$ and $N = 200$, the fixed- q approximation gives $\beta_c \approx 2.47$, $\beta_t \approx 3.75$ and $\beta_{\text{bal}} = 5$. A fitted value can lie above the first two and still below the third. In that theoretical case, an ordered state is preferred and completion may be possible once it is reached, but a population starting near balance still has to escape its locally stable starting state.

2.4 Effective Potts Parameter

The fitted value $\hat{\beta}$ summarizes a complete measured response curve. H2 also asks whether the same parameter describes every part of that curve. The effective parameter $\beta_{\text{eff}}(s)$ provides a state-by-state check by asking which value of β would reproduce the adoption probability at one leader share.

In the controlled states, the remaining support is divided as evenly as possible among the other $q - 1$ opinions. Ignoring small differences caused by integer counts, each non-leading opinion has share $(1 - s)/(q - 1)$. The Potts leader-adoption probability is then

$$P_{\text{lead}}(s) = \frac{e^{2\beta s}}{e^{2\beta s} + (q - 1)e^{2\beta \frac{1-s}{q-1}}}.$$

Write this probability as p . Dividing it by the probability $1 - p$ of choosing another opinion cancels the common denominator. Multiplying the resulting odds by $q - 1$ and taking the logarithm gives

$$\frac{p}{1-p} = \frac{e^{2\beta\left(s - \frac{1-s}{q-1}\right)}}{q-1},$$

$$\ln\left(\frac{p(q-1)}{1-p}\right) = 2\beta\left(s - \frac{1-s}{q-1}\right).$$

Solving for β defines the effective Potts parameter,

$$\beta_{\text{eff}}(s) = \frac{\ln\left[\frac{P_{\text{lead}}(s)(q-1)}{1-P_{\text{lead}}(s)}\right]}{2\left(s - \frac{1-s}{q-1}\right)}.$$

If one constant β describes the measured states, these effective values should agree with one another and with $\hat{\beta}$, within sampling uncertainty and the approximation introduced by integer counts. Systematic variation indicates that one constant does not reproduce the response under this Potts specification. It does not establish that an underlying tendency to conform actually changes.

Under the symmetric Potts response, both the numerator and denominator of the inversion are zero at exact balance. The balanced point alone therefore contains no information about β , which is why measurements at small positive leads are needed. The effective parameter depends on the Potts functional form, whereas the drift compares the measured adoption probability directly with the current share. Both remain conditional on the states and presentation used in the controlled experiment. The following chapter describes how those states are constructed, how responses are measured and fitted, and how the resulting quantities are compared with opinion dynamics simulations.

Chapter 3

Methods

3.1 Models and Setup

The empirical analysis consists of two types of experiments. In the opinion dynamics simulations, agents repeatedly update their opinions. Each update can change the population and with that the opinion distribution shown to the next agent. In the controlled response estimation, the opinion distribution is instead fixed in advance. Each controlled state is presented repeatedly to estimate how often the model selects each opinion. Both experiments use the same models and prompt format. Their main difference is whether the displayed distribution results from earlier interactions or is set by the experimental design. This chapter introduces the setup shared by both experiments, after which the two experiment types are described separately.

The study includes six open-weight models from three model families, with a smaller and a larger model from each family. Open-weight models make their parameters available for download, which allows a fixed model state to be run on local hardware throughout the study. Instruction-tuned models have received further training to follow natural-language requests which makes them suitable for the decision-task prompt used in the experiments.

The selection supports multiple comparisons. Comparing model families shows whether an observed behavior occurs across different LLMs. Comparing two sizes within each family makes it possible to check whether the results are consistent with the relationship between model capability and conformity strength reported in the binary setting (De Marzo, Castellano, et al., 2026) and look for similar tendencies of model capacity with the additional factors introduced in the thesis. As mentioned before, Open-weight models can be run locally, so organizations and individuals keep

their data under their own control and do not depend on a single provider (Kapoor et al., 2024). For this study this means that every query was answered by the same model state and that the checkpoints stay available for replication, which a model accessed through an application programming interface (API) cannot guarantee. Qwen and Llama are the two most downloaded open-weight families on Hugging Face, and Gemma accounts for about a tenth of the derivative models uploaded there (Lambert & Brand, 2026). All three models come from different developers. Llama also connects this study to the binary experiments, which used Llama 3 70B. The models are listed in the following table.

The size of the larger model in each family was limited by what fits on one graphics processing unit (GPU) with 80 GB of memory and room for inference. For this reason the largest model, Llama 3 70B, was run as a quantized checkpoint, produced with 4-bit Activation-aware Weight Quantization (AWQ) (J. Lin et al., 2024). Its weights are stored using 4-bit instead of 16-bit numerical precision, reducing the memory required to run it. Because quantization may change the model’s responses (Dutta et al., 2024), its results are interpreted as results of the quantized checkpoint, and comparisons with published full-precision results are made cautiously.

TABLE 3.1: List of models that were the subjects of the analysis

Family	Model	Precision
Llama	Llama 3.1 8B	full
Llama	Llama 3 70B	4-bit AWQ
Qwen	Qwen 2.5 7B	full
Qwen	Qwen 2.5 32B	full
Gemma	Gemma 4 E4B	full
Gemma	Gemma 4 31B	full

Checkpoints were obtained from Hugging Face, using the vLLM inference engine (Kwon et al., 2023), primarily on a single NVIDIA H100 GPU with 80 GB of memory. Models were loaded one at a time so that the checkpoint and the memory required during inference fit in GPU memory. This made the more than one million required model queries feasible without relying on external API availability, sufficient funds for credits or running into rate limits.

Temperature was set to 0.2 throughout the experiments. At this relatively low value, the model strongly favors its most likely outputs, so repeated queries usually produce consistent responses. Nucleus sampling was set to $p = 0.9$, which further limits unlikely responses. Independent queries and independent simulation runs were processed in batches where possible. The estimated adoption probabilities describe model behavior under these fixed settings. Because replies are sampled at a positive temperature and batched GPU inference is not bitwise deterministic (Yuan et al., 2025), repeating the data collection can produce different individual replies and different estimated frequencies.

Every query follows the same format. The prompt shows the focal agent a list of all other agents, along with the opinion each currently supports, one agent per line, and asks it to reply with the opinion it wants to support, enclosed in square brackets, matching the prompt design from the binary study (De Marzo, Castellano, et al., 2026). The focal agent’s own entry is omitted, and each query contains no history of earlier interactions. This implements the memoryless setup of Chapter 2: the agent receives neither its previous opinion nor earlier decisions. Its response can still depend on the wording and labels in the current prompt. Agent names are random three-character strings generated for every query, and opinions are neutral labels, arbitrary two-letter strings such as aa and ab. Because LLMs can be sensitive to the identity and position of answer options (Pezeshkpour & Hruschka, 2024; Zheng et al., 2024), randomizations were applied to every prompt. The mapping between internal opinions and displayed labels was randomly permuted for every query, and the order of agents in the list was shuffled. Thus, no opinion was linked to a particular label or position in the prompt. It does not make the labels neutral in the sense that the models are indifferent to them. Each internal opinion is equally likely to receive a preferred label, but the measured response still averages over label preferences and remains conditional on the label pool. An example of the resulting prompt is shown below.

Below you can see the list of all the other AI agents with the opinion they support.

You must reply with the opinion you want to support.

The opinion must be reported between square brackets.

fYQ: ay

aGd: bk

CI5: ae

...

Reply only with the opinion you want to support, between square brackets.

Replies are parsed by extracting the label between square brackets, with a fallback that accepts a reply if it contains exactly one valid label. Responses that cannot be matched to an option are flagged as invalid and excluded from the controlled response estimates. In the opinion dynamics simulations, an invalid reply leaves the focal agent's opinion unchanged, but the update still counts toward elapsed time. Valid-response rates exceeded 99 percent in the main neutral-label experiments.

3.2 Opinion Dynamics Simulations

This section describes the first of the two experiments. The multi-option opinion dynamics simulations that produce the collective behavior analyzed in the descriptive part of Chapter 4. The guiding question follows directly from the research questions. When a population begins with q opinions, does one opinion emerge as the leader? If so, how quickly does this happen, and does its support grow toward consensus or settle at a partially coordinated state? The simulations examine how these outcomes change with the number of opinions q , the group size N , and the model used. Every simulation starts at the balanced state. The q opinions are distributed as evenly as the integer population allows, so the initial counts differ by at most one agent.

A run then repeats the agent update of Section 3.1. One agent is selected at random, shown the current opinions of all other agents with fresh random names and

altered opinion labels. Its opinion is replaced by the option selected in a valid reply. One time step or sweep corresponds to N such updates, so after one time step t every agent has been updated once on average. The complete distribution across all opinions was stored at each recorded time point, approximately ten times per sweep. This distribution was later used to calculate the metrics for analysis reported in Chapter 4 such as the leader share $s(t)$, the coordination level $C(t)$, and the number of active opinions $K(t)$.

Consensus was treated as the end of a run. Once all agents supported the same opinion, no further updates were performed and the exact consensus time was recorded. An opinion with no remaining supporters disappeared from the displayed list, although its label remained valid for the parser and could reappear. Runs last at most 100 time steps, that is, $100N$ individual updates.

The core simulation combinations covered $N \in \{25, 50, 100, 200\}$ and $q \in \{3, 10, 50\}$ for all six models. The design required every opinion to be represented by at least one agent at the beginning of a simulation, so only conditions satisfying $q \leq N$ were possible; when $q = N$, each opinion initially had exactly one supporter. For the Gemma and Qwen models, the population range was extended to $N \in \{400, 800\}$ at $q \in \{3, 10, 50\}$. A supplementary condition used Qwen 2.5 32B at $q = 100$ and $N = 200$. Every condition was simulated with a minimum of three runs. Runs were parallelized for efficiency as described in Section 3.1, while the steps within one run always remained sequential.

Following the approach of the binary study (De Marzo, Castellano, et al., 2026), simulation outcomes are compared across conditions and interpreted using the theoretical benchmarks of Chapter 2. No regression was fitted. The relationships with group size and the number of opinions vary between models, so a pooled estimate could hide these differences. Separate estimates for each model would also be limited by the small number of runs in most conditions.

3.3 Controlled Experiments

The second experiment measures how an individual model responds to a specific opinion distribution. At each controlled state, the same condition is presented repeatedly, and the proportion of responses selecting the leading opinion provides an

estimate of $P_{\text{lead}}(s)$, as in the controlled-state measurement of the bias study (De Marzo, Bellina, et al., 2026). These measurements connect the individual choices to the conceptual framework of Chapter 2. They are used to estimate the conformity strength β , compare it with the theoretical benchmarks, and calculate the drift $P_{\text{lead}}(s) - s$.

A possible approach would be to estimate these probabilities from the choices made during the opinion dynamics simulations. However, the opinion distributions observed during a simulation depend on all previous choices. Models that rapidly approach full coordination pass quickly through intermediate states; other models may remain near balance or at an intermediate state. Because of this, some opinion distributions are observed only a few times or not at all, and different models are not evaluated at the same states. Additionally, when an opinion loses all its supporters, states in which it remains active are rarely observed again. Thus, observations concentrate around the states each model visits most often. Other theoretically relevant states may be poorly represented or unobserved. The controlled experiment avoids this problem by defining the opinion distributions in advance and presenting the same states repeatedly to every model.

That queries in the controlled experiment not being dependent on earlier decisions by agents also has a practical side. Prompts can be processed in large parallel batches. The same hardware therefore produces far more controlled observations than dynamic observations in the same time. The controlled states can also be concentrated in theoretically important regions, for example, close to balance. Across all models, values of q , and displayed group sizes, the controlled design analyzed in Chapter 4 comprises 254,023 valid responses.

Controlled experiments also have shortcomings in the multi-option setting that must be acknowledged. The state space is simpler in the binary case. When $q = 2$, knowing the number of supporters of one opinion also determines the number supporting the other. For example, in a population of $N = 200$, if the first opinion has 70 supporters, the second must have 130. The first count can take any integer value from 0 to 200, giving 201 possible count distributions in total. Each binary state can therefore be described by the single collective opinion m .

With more than two opinions, knowing one opinion’s share is no longer sufficient to describe the state because the remaining support can be divided among the other opinions in many different ways. Since all q shares must sum to one, $q - 1$ of them

can vary independently. The resulting space of possible opinion-share distributions is mathematically called a $(q - 1)$ -dimensional simplex. Its dimensionality expands with q , while the number of possible count distributions inside it grows rapidly with both N and q . At $N = 200$ and $q = 10$, for instance, there are more than 10^{15} possible labeled count distributions. A feasible sample would cover only a small part of this space.

The experiment instead uses a one-dimensional slice in which the leader’s share varies while the remaining support is distributed as evenly as possible. This slice extends from near balance toward a strong lead while keeping every opinion represented. Because every other opinion keeps at least one supporter, the largest displayable leader share is $s_{\max} = (N - q)/(N - 1)$, which decreases with q at fixed N . It keeps the competing opinions approximately evenly matched and permits direct comparisons across models and values of q . Its limitation is that it excludes configurations with a strong runner-up or several strong competitors. The measured $P_{\text{lead}}(s)$ thus cannot be assumed to describe the entire space of opinion distributions. Appendix A tests the fitted softmax on eight supplementary states, including an even-rest control and uneven distributions, while Chapter 4 compares predictions from the measured responses with the opinion dynamics simulations.

3.3.1 Design of the Controlled States

Each measurement block fixes one model, one number of opinions q , and one displayed group size. It measures a sequence of controlled states from near balance toward larger leader shares. In each state, one opinion is designated as the leader, while the remaining support is divided as evenly as possible among the other $q - 1$ opinions. The displayed group contains $N - 1$ agents because the focal agent is excluded from its own prompt.

The leader count is determined by a design parameter $r \in [0, 1]$:

$$\begin{aligned} c_{\text{lead}}(r) &= \text{round}[c_{\min} + r(c_{\max} - c_{\min})], \\ c_{\min} &= \left\lceil \frac{N - 1}{q} \right\rceil, \\ c_{\max} &= (N - 1) - (q - 1). \end{aligned}$$

Here, $c_{\min}$ is the leader count closest to equal support, while $c_{\max}$ is the largest count that still leaves at least one supporter for every other opinion. The parameter r is a relative position between these two counts. The point $r = 0$ represents the closest feasible state to balance, while $r = 1$ gives the largest leader count for which every competing opinion remains with one supporter. As an illustration, with $q = 5$, 50 agents (49 displayed), and $r = 0.4$, the leader receives 24 supporters and the remaining counts are 7, 6, 6, 6.

For $q \geq 3$, the core design uses the following r grid:

$$r \in \{0, 0.03, 0.06, 0.1, 0.15, 0.20, 0.25, 0.4, 0.6, 0.8, 1.0\}.$$

Duplicate count distributions produced by rounding are omitted. Each state in the main neutral design was presented 150 times.

The concentration of points at small r provides more detail close to balance. This is where the response is expected to provide the most information about conformity strength. At larger leads, strongly conformist models may select the leader in almost every query. For these models, the states at very small leads, where the response is still visibly probabilistic, provide the information needed to estimate β .

3.3.2 Fitting the Response Function

For the analysis, each combination of model, number of opinions q , and displayed group size is treated separately. At every controlled state, it was recorded how many valid responses were collected and how often each option was chosen. These counts are used to estimate the conformity strength $\hat{\beta}$, quantify its sampling uncertainty, and assess how closely the fitted Potts softmax describes the measured response.

The conformity strength is estimated by maximum likelihood. For any candidate value of β , the Potts softmax assigns an adoption probability to every option at every controlled state. Let j identify a controlled state and let $n_{j\alpha}$ be the number of times option α was chosen at that point. The log-likelihood is

$$\ell(\beta) = \sum_j \sum_{\alpha=1}^q n_{j\alpha} \log P_{\beta}(\alpha \mid m_j).$$

The estimate $\hat{\beta}$ is the value that maximizes this expression. It is the conformity strength under which the observed pattern of choices is most likely according to the Potts softmax. Using the log-likelihood prevents the product of thousands of probabilities from becoming too small for reliable numerical representation. The logarithm converts this product into a sum while leaving the maximum at the same value of β .

Because the estimate is based on a finite number of queries, a repetition of the experiment would not produce exactly the same adoption frequencies or fitted value. This sampling uncertainty is estimated using 200 bootstrap resamples (Efron, 1979). Within each combination of model, q , and displayed group size, responses are sampled with replacement. The Potts softmax is fitted again to every resample. The 2.5th and 97.5th percentiles of the resulting values form the reported 95 percent bootstrap interval.

Each fitted value is compared with the critical value $\beta_c(q)$ for the same number of opinions. This comparison examines the equilibrium argument motivating H1 alongside the observed completion and consensus times. The effective values $\beta_{\text{eff}}(s)$ provide a state-by-state check. They are calculated at individual controlled states by inserting the measured leader-adoption probability into the inversion derived in Section 2.4. The leader-adoption proportion enters the inversion with the Haldane-Anscombe correction as $(k + 0.5)/(n + 1)$ (Anscombe, 1956; Haldane, 1956), where k is the number of queries in which the leader was chosen and n the number of valid queries at the state. The correction is applied at every state; it changes the interior values only marginally and keeps the logarithm finite where the leader was chosen in all or none of the queries. Its 95 percent interval is an approximate Wald interval on the logit scale, transformed through the same inversion. States in which the leader was chosen in all or none of the queries are omitted from the effective-parameter plots in Chapter 4. The correction makes their values finite but does not make them precise estimates.

Estimating $\hat{\beta}$ does not by itself show how well the Potts softmax matches the measured response. Its goodness of fit is checked by comparing the predicted leader-adoption probabilities with the measured frequencies at each controlled state. These differences are summarized by the query-weighted root mean squared error (RMSE),

$$\text{RMSE} = \sqrt{\frac{\sum_j n_j (\hat{p}_{\text{lead},j} - p_{\hat{\beta},j})^2}{\sum_j n_j}},$$

where $\hat{p}_{\text{lead},j}$ is the measured leader-adoption frequency, $p_{\hat{\beta},j}$ is the probability predicted by the fitted softmax, and n_j is the number of valid queries at controlled state j . Squaring prevents positive and negative differences from canceling and gives greater weight to larger deviations. Taking the square root returns the result to the original probability scale. As an example, a RMSE of 0.03 corresponds to an error scale of about three percentage points. A small RMSE indicates that the fitted curve follows the measured leader-adoption probabilities closely. A large value indicates that one constant β does not reproduce the full response curve adequately.

The preceding sections described the design and analysis of the opinion dynamics simulations and controlled measurements. Before applying this approach to the multi-option setting, the following section examines whether the design recovers the central binary patterns reported by De Marzo, Castellano, et al. (2026).

3.4 Replicating the Binary Experiment

To check the opinion dynamics simulation, the binary experiment reported by De Marzo, Castellano, et al. (2026) was repeated with the Llama 3 70B model. Each update used the opinion labels k and z , randomly swapped their assignment, and used the “friends” wording of the original prompt.¹ The population contained $N = 50$ agents, and each of the twenty runs lasted ten sweeps.

¹The replication prompt states: “Below you can see the list of all your friends together with the opinion they support.” The remaining experiments use the formulation: “Below you can see the list of all the other AI agents with the opinion they support.” This follows the AI-agent wording of De Marzo, Bellina, et al. (2026).

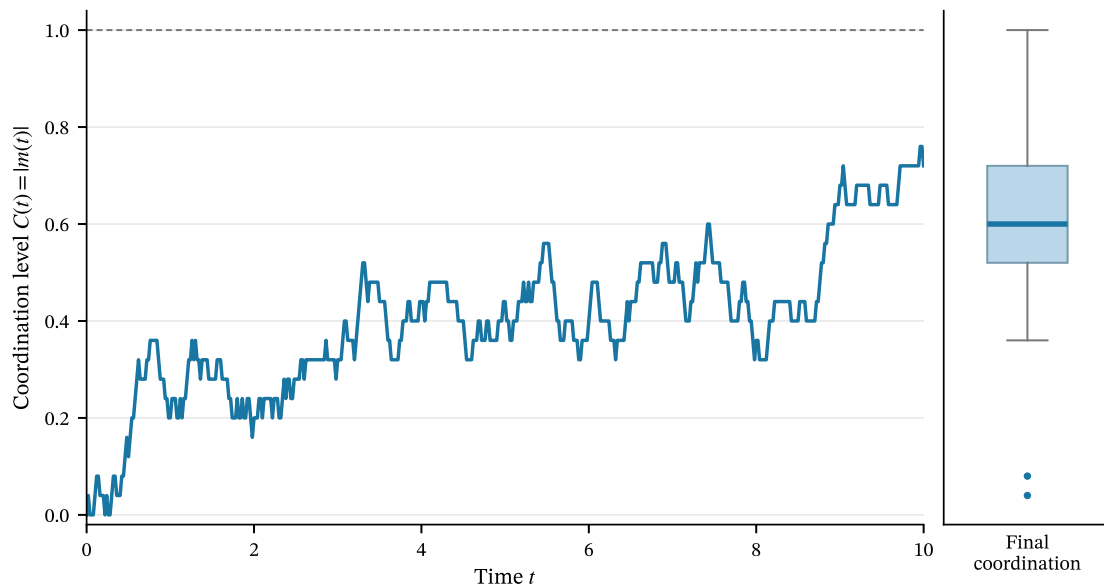


FIGURE 3.1: Binary Llama 3 70B replication. The chart displays the run closest to the mean of all runs across twenty simulations. The box plot summarizes the final coordination level.

The replication recovers the behavior reported by De Marzo, Castellano, et al. (2026): Llama 3 70B develops substantial coordination but often stalls below consensus within the investigated time frame. The displayed trajectory reaches a similar coordination level, although the final levels across all twenty runs are lower than in the binary study. That study used a Replicate-hosted model via API,² whereas this replication used a locally hosted 4-bit AWQ model. These differences may have contributed to the lower values.

The first replication checks the opinion dynamics simulation. A second check compares the controlled response estimation against the published binary response function. For $q = 2$, the leader share s of a state can be converted to the binary collective opinion via

$$|m| = 2s - 1.$$

²The code and data of the binary study are published at <https://github.com/giordano-demarzo/LLMs-Opinion-Dynamics> and archived on Zenodo (doi:10.5281/zenodo.17911594), as stated in the data availability statement of De Marzo, Castellano, et al. (2026).

For example, a leader share of $s = 0.7$ corresponds to $|m| = 0.4$. In the figure, m is signed according to the support for the label, and $P(m)$ is the observed probability of choosing that label. Randomized label assignments place the label on both the larger and the smaller side which leads to measurements at positive and negative m . This gives the same response quantity used in the binary framework of De Marzo, Castellano, et al. (2026). As shown in Section 2.2, the Potts response rule reduces to the binary hyperbolic-tangent response when $q = 2$. Consequently, the fitted β from the $q = 2$ data can be compared directly with the published binary response curves.

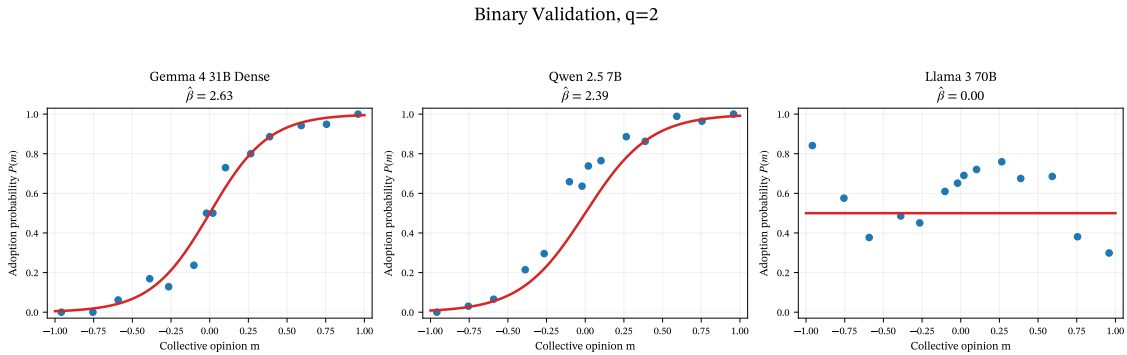


FIGURE 3.2: Binary validation at $q = 2$ and a group size of $N=50$ agents, one model per family. Points show the adoption probability at each controlled state. Curves show the binary response function with the fitted conformity strength. Gemma 4 31B and Qwen 2.5 7B follow the tanh form with $\hat{\beta} = 2.63$ and 2.39 . The Llama 3 70B response is not of tanh form, and its fitted value collapses to zero.

The second figure shows this comparison for one member of each model family. For Gemma 4 31B and Qwen 2.5 7B, the measured points broadly follow the tanh shape. Their fitted conformity strengths of 2.63 and 2.39 are also of the same magnitude as the values reported for capable models in the binary study (De Marzo, Castellano, et al., 2026). Both estimates lie above the fluctuation threshold $\beta_t(50) \approx 1.96$ introduced in Section 2.1. The binary framework places these models in the coordinated regime at $N = 50$, where full coordination is expected on ordinary timescales. The controlled estimation replication recovers the binary response function for these models.

The third plot is different. The Llama 3 70B response at $q = 2$ does not follow a tanh shape. Adoption initially increases with the size of the majority but declines again

when that majority becomes extreme. As a result, the fitted parameter approaches zero, which is a sign of the poor fit. Similar departures from the tanh shape were documented in the binary study. The prompt analysis of the binary study (Supplementary Fig. 2) shows that for Llama 3 70B two of five prompt variants produce adoption curves that a tanh function does not describe well (De Marzo, Castellano, et al., 2026). The present estimation uses the AI-agent wording, a different label set to represent more opinions, as well as a locally hosted quantized model.

The two checks are generally consistent with the published binary results. The dynamics replication recovers the reported pattern of partial coordination and stalling, while the controlled estimation recovers the tanh response for 2 of the 3 models. Where the Llama 3 70B response differs, its poor fit is consistent with previously documented deviations from the tanh-shaped response for this model. These comparisons are intended as consistency checks instead of exact reproductions. Differences in hosting and precision due to quantization and for the response estimation prompt wording and labels limit the comparison with the original setup. Even so, the checks show that the methodological design can reproduce the main patterns of the binary setting and provide a reference point for the multi-option analysis.

Chapter 4

Results

4.1 Collective Coordination Dynamics

Figure 4.1 gives a first overview of the opinion dynamics behavior. Each plot shows the coordination level $C(t)$ at $N = 100$ for an initial number of opinions.

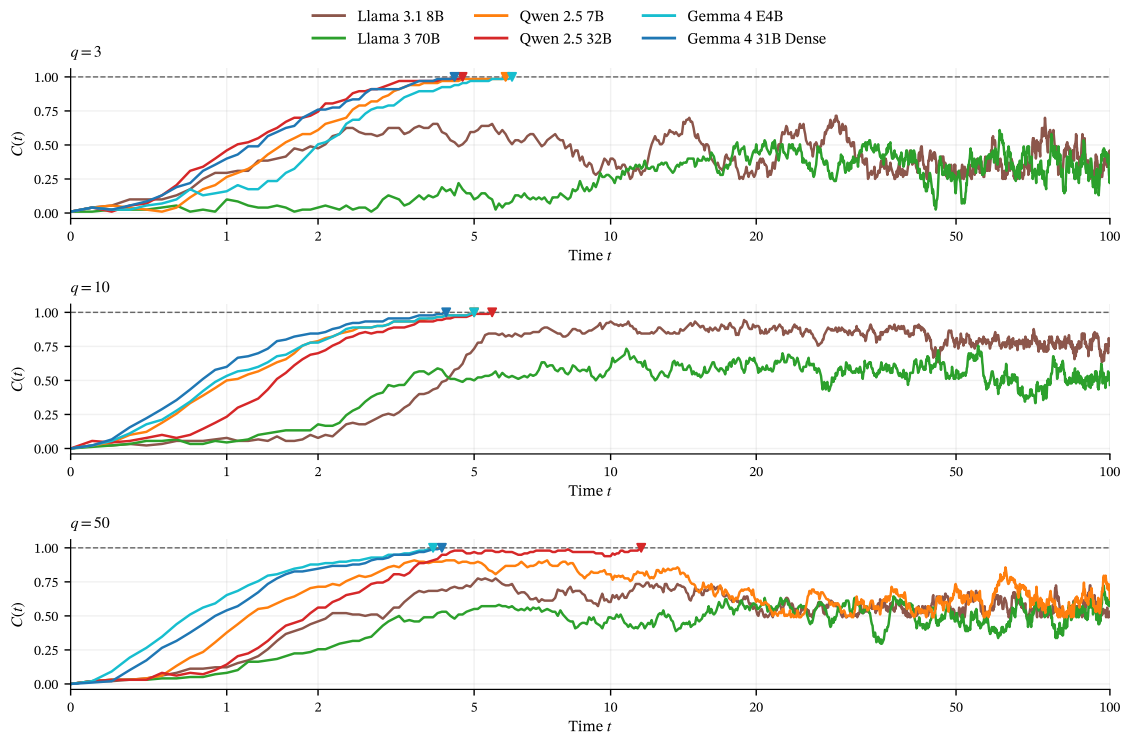


FIGURE 4.1: Trajectories of the coordination level $C(t)$ at $N = 100$. Rows show the initial number of opinions and colors the models. For each condition the run whose final value lies closest to the median across all available runs is shown. Time uses a shifted logarithmic scale. Triangles mark the time of consensus.

The displayed outcomes differ more strongly between models than between numbers of opinions. The two Gemma models and Qwen 2.5 32B reach consensus in every displayed condition within roughly three to ten time steps, and their trajectories at $q = 50$ look very similar to their trajectories at $q = 3$. Qwen 2.5 7B follows the same pattern up to $q = 10$ but stalls at $q = 50$, where its coordination level peaks and then falls back to an intermediate plateau. The two Llama models do not reach consensus at this group size at any q ; their trajectories rise to partial coordination and remain there for the rest of the horizon.

A leading opinion separates from the field within the first few sweeps, visible as the steep early rise on the log time axis. Increasing q does not produce a uniform change in the displayed outcomes. Still, the Qwen 2.5 7B results show that additional opinions can change whether full coordination is reached within the observation horizon. The following sections examine the emergence of a plurality, the extinction of opinions and the dependence of consensus time on group size.

Where Figure 4.1 compared conditions through the rescaled coordination level, Figure 4.2 shows the raw leader share $s(t)$. The balanced starting points are no longer aligned. The curves begin at $1/q$, so the trajectories for $q = 3, 10$ and 50 have different start points according to the initial share in the balanced state. The figure shows how quickly this gap closes.

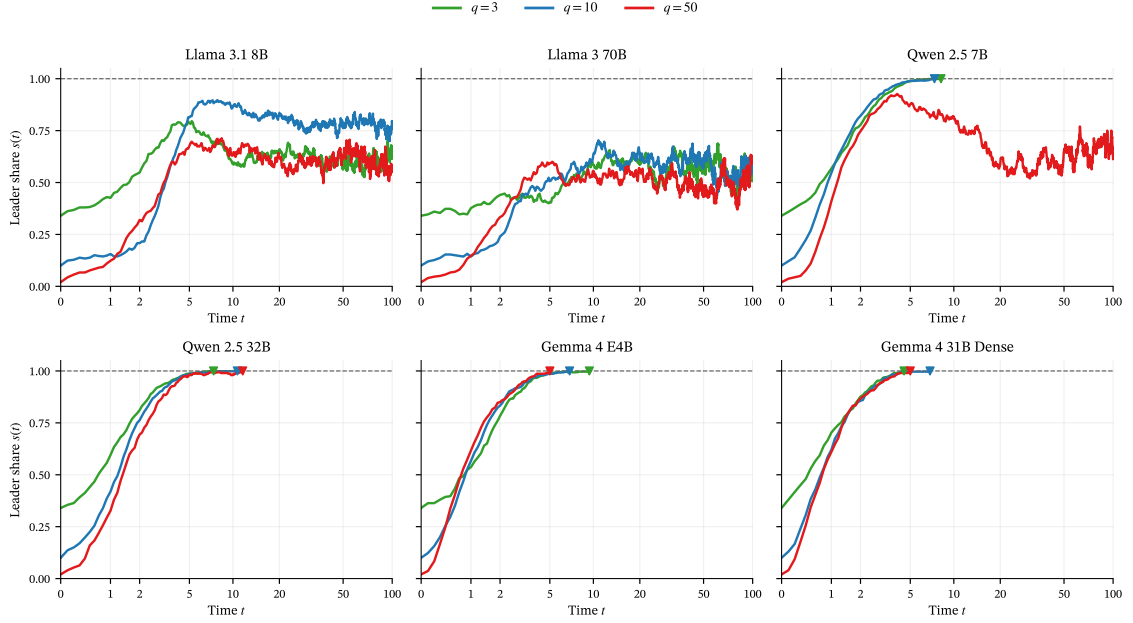


FIGURE 4.2: Mean leader-share trajectories $s(t)$ at $N = 100$, averaged over all available runs in each condition. Time uses a shifted logarithmic scale, $\log(1 + t)$, to resolve the rapid early formation of a plurality within the full horizon. Triangles mark the time at which the mean trajectory reaches consensus.

For the conditions that reach consensus the steep rise happens in the first two to three time steps, and the mean trajectories reach consensus after about five to twelve time steps with no systematic ordering by the number of opinions. At $q = 50$ the emerging leader multiplies its initial support roughly fortyfold within a few time steps. The number of opinions leads to no uniform time cost: from $q = 3$ to $q = 10$ the mean trajectory reaches consensus earlier for Gemma 4 E4B and Qwen 2.5 7B, and for the Gemma models the difference between three and fifty opinions stays within a few time steps either way. Where a cost appears, it appears abruptly. For Qwen 2.5 32B, the mean trajectory reaches consensus after approximately 7.4 time steps at $q = 3$ and 11.6 at $q = 50$, and Qwen 2.5 7B goes from a shortened consensus time to the stall at $q = 50$. The conditions without consensus fail late. Qwen 2.5 7B at $q = 50$ forms a plurality as fast as the populations that reach consensus and reaches a mean leader share of 0.93 after four time steps before reversing to 0.61. The mean leader share of Llama 3.1 8B peaks at approximately 0.80 within the first nine time steps, approaching 0.9 at $q = 10$ without ever completing. Llama 3 70B is noticeably slower in the emergence stage itself. On the whole, strong

pluralities generally form within the first few sweeps, without a uniform slowdown as q increases.

Figure 4.3 shows the decline in the number of active opinions $K(t)$.

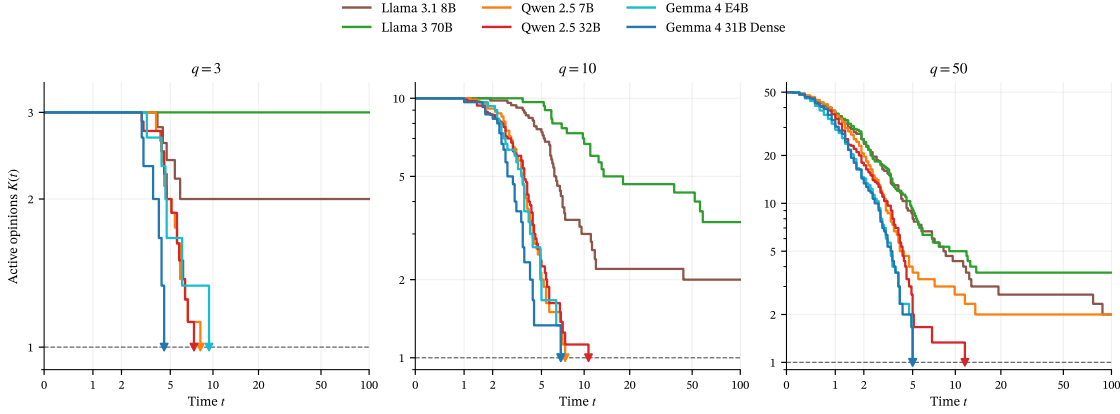


FIGURE 4.3: Mean number of active opinions over time at $N = 100$, averaged over all available runs in each condition. Panels show the initial number of opinions and colors the models. Time uses a shifted logarithmic scale and the number of active opinions a logarithmic scale. Triangles mark conditions that reach a single remaining opinion (full coordination).

Opinion extinction proceeds rapidly across all models. Starting from $q = 50$, the populations lose approximately half of their active opinions within about two sweeps, including conditions that never reach consensus. The coordinating models continue this process without interruption, reaching two active opinions after approximately four to five time steps before the extinction of the last rival opinion. In the conditions without consensus, extinction instead slows or stops near the end. Qwen 2.5 7B remains with two opinions at $q = 50$, and Llama 3.1 8B ends with two opinions in every displayed condition. Only Llama 3 70B retains three to four active opinions at the end of the observation horizon. The later parts of the Llama trajectories show that extinction continues during the plateau phase regardless. At $q = 10$ and $q = 50$, for example, Llama 3.1 8B reaches its final pair only after approximately forty to ninety steps. Due to this, the initial number of opinions determines the number of opinions in play only during the early part of the dynamics. After that, every population gets reduced to a contest among a small number of surviving opinions. The prolonged endgame of the stalled conditions consequently occurs in an effectively binary or three-opinion setting rather than in the initial q -opinion

configuration. How this reduction changes the applicability of the fixed- q Potts description and the experimental design of this thesis is examined in Section 4.3 and interpreted further in Chapter 5.

Figure 4.4 completes the descriptive picture by following the time to consensus across group sizes.

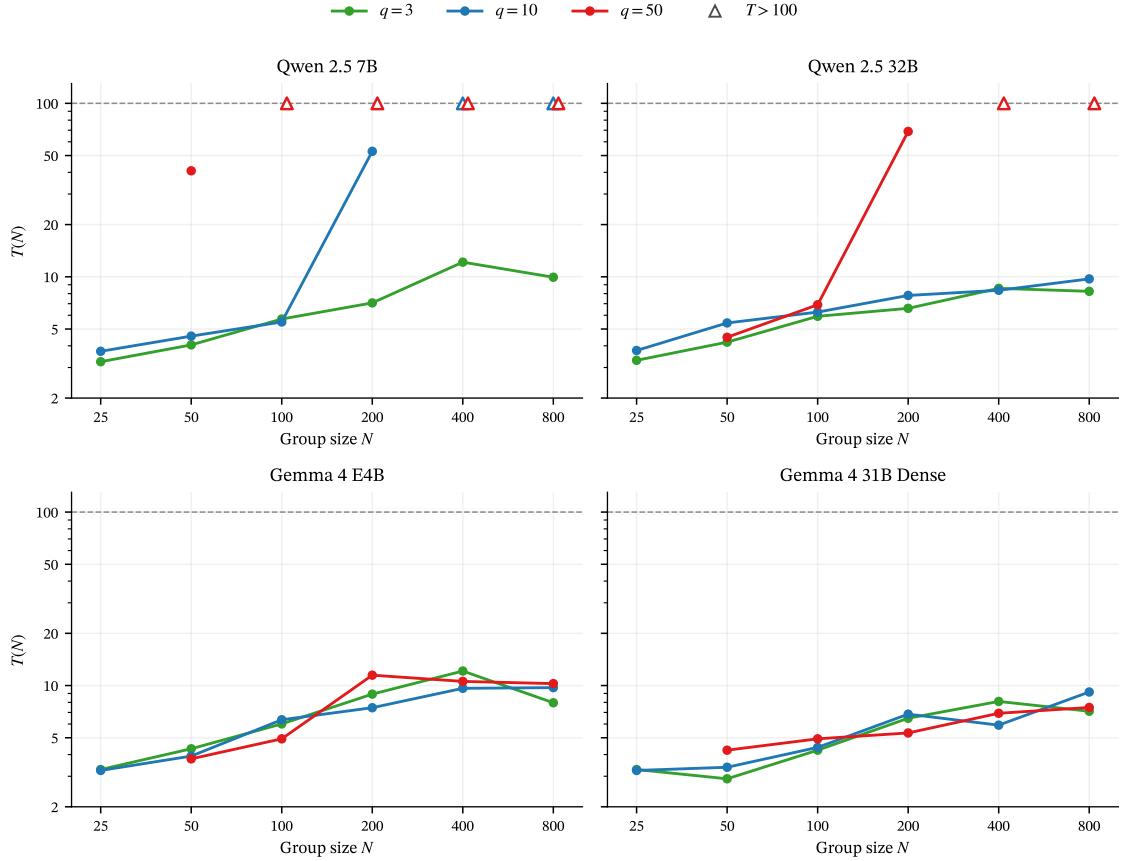


FIGURE 4.4: Consensus time against group size for the four models with observable consensus times. Each point is the median over the runs that reach consensus; where only part of the runs finish, the point refers to the completing runs. Open triangles mark conditions in which no run reaches consensus within the horizon of 100 time steps. Colors distinguish the initial number of opinions. Both axes are logarithmic.

The differences between models are clear. Gemma 4 31B reaches consensus in every displayed condition, with median times increasing from roughly three time steps at $N = 25$ to between seven and nine steps at $N = 800$. The $q = 50$ condition is among the fastest at the largest group sizes. Gemma 4 E4B also reaches consensus

throughout all combinations, although its median times vary less smoothly and reach approximately 12.1 steps. At $N = 800$, all three values remain between approximately eight and ten steps. Qwen 2.5 32B shows a similar moderate increase at $q = 3$ and $q = 10$.

For the Qwen models, increasing group size can turn rapid consensus into prolonged disagreement. At $q = 50$, Qwen 2.5 32B completes within a few time steps at $N = 50$ and 100, but no run finishes within 100 time steps at $N \geq 400$. At $N = 200$, its median consensus time rises from about eight time steps at $q = 10$ to 51 at $q = 100$. All three runs finish at $q = 100$, compared with two of three at $q = 50$, so completion does not decrease at every increase in q . Qwen 2.5 7B encounters difficulty at smaller populations and reaches consensus at $q = 50$ only when $N = 50$. The loss of reliable completion as populations grow resembles the critical group size behavior in the binary framework, with the change also depending on the model and number of competing opinions.

4.2 Measured Response Functions and Potts Fits

The previous section described the collective opinion dynamics. This section examines the individual choices that produce those dynamics. It presents the controlled measurements introduced in Chapter 3, in which agents respond to controlled opinion distributions. At each state, one opinion leads while the remaining support is divided as evenly as possible among the other opinions. Repeating each query provides an estimate of the probability that an agent adopts the leading opinion.

The Potts softmax is fitted separately for each model, number of opinions and group size. Each fit summarizes the measured response with one conformity strength $\hat{\beta}$. Comparing $\hat{\beta}$ with the critical value $\beta_c(q)$ indicates whether the fitted Potts model favors an ordered state over balance. The fluctuation threshold provides a further benchmark for the final approach to consensus once an ordered state has formed. Together, these comparisons help connect the measured choices of individual agents with the collective outcomes reported in the previous section.

Figure 4.5 shows the measured response functions at $N = 200$. Each graph represents one model, while colors distinguish the number of opinions. The points show the measured leader-adoption frequencies, and the solid lines show the fitted Potts response curves.

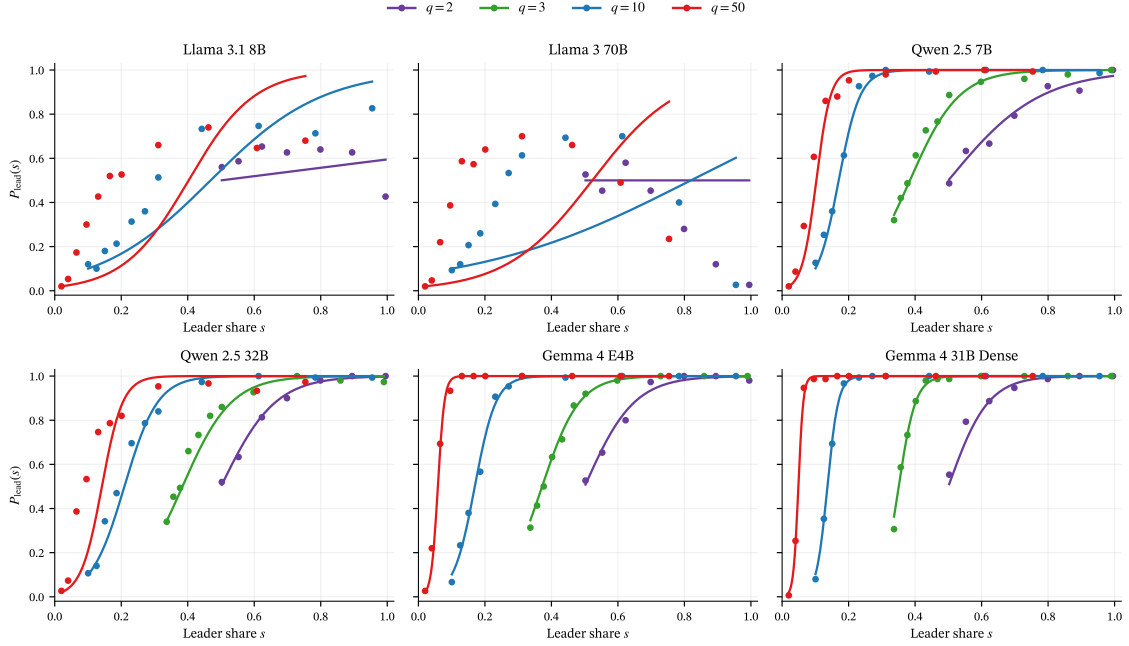


FIGURE 4.5: Measured leader-adoption probabilities and fitted Potts response curves at $N = 200$. Points show the measured frequencies at the controlled states and colored curves the fitted Potts softmax.

The smaller group sizes show the same ordering of the models.

For the Gemma and Qwen models, the fitted curves follow the measured responses reasonably closely. Starting from the chance level $1/q$ at balance, the probability of choosing the leader increases rapidly and approaches one after a relatively small lead has formed. The weighted RMSE remains below 0.03 for both Gemma models. It is higher for the Qwen models, reaching 0.07 for Qwen 2.5 7B and 0.10 for Qwen 2.5 32B. The RMSE only measures agreement between the fitted curve and the controlled states included in the experiment. It does not show whether the fitted response also applies to other opinion distributions or predicts the population dynamics correctly. These questions are examined in Section 4.3.

Fits of the Llama models are less consistent with the Potts softmax. Their leader-adoption probabilities increase more slowly and remain below one over most of the measured range. For Llama 3 70B, the probability even declines again at larger leader shares which means that the model follows extreme majorities less often than moderate ones. The corresponding RMSE values of 0.17 and 0.27 show that a single fitted $\hat{\beta}$ does not reproduce the shape of these response curves well. Their collective

behavior is instead analyzed in Section 4.3, where the measured probabilities are used directly.

Figure 4.6 collects the fitted conformity strengths of all blocks and places them against the critical value $\beta_c(q)$, with one plot per group size.

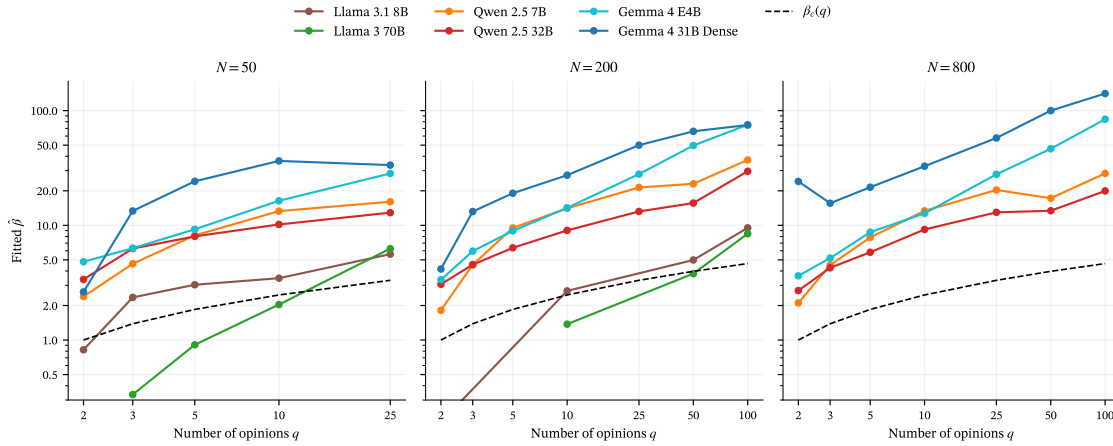


FIGURE 4.6: Fitted conformity strength $\hat{\beta}$ against the number of opinions on logarithmic axes. Each graph shows one group size and colors the models. The dashed curve is the critical value $\beta_c(q)$. The Llama estimates are included for completeness, but should be interpreted with caution because their measured response curves deviate substantially from the shape assumed by the Potts softmax.

For the four models whose responses are described well by the Potts softmax, every fitted value lies above the critical value $\beta_c(q)$ across all numbers of opinions and group sizes. Thus, within the fitted Potts model, an ordered state is preferred over balance in every condition. This is consistent with the formation of leading opinions in the simulations.

The fitted values exceed $\beta_c(q)$ by factors ranging from approximately two to twenty. This difference generally becomes larger as q increases because the fitted conformity strength rises faster than the critical value. Hence, the difficulty caused by additional opinions cannot be explained by these models falling below $\beta_c(q)$. In some conditions, the model chooses the leading opinion almost every time, even at the smallest measured lead. This leaves little variation to determine β precisely, so these fits are treated as poorly determined estimates. Their reported 95 percent bootstrap intervals still remain above $\beta_c(q)$.

The graphs also show different relationships between the fitted parameter and group size. The estimates for Gemma 4 31B generally increase with group size, whereas those for Gemma 4 E4B and the two Qwen models remain comparatively stable. The fitted Llama values decrease. The figures below examine the dependence on group size more directly.

The binary blocks allow a direct comparison with the framework of De Marzo, Castellano, et al. (2026). Figure 4.7 shows the fitted conformity strength at $q = 2$ as a function of group size.

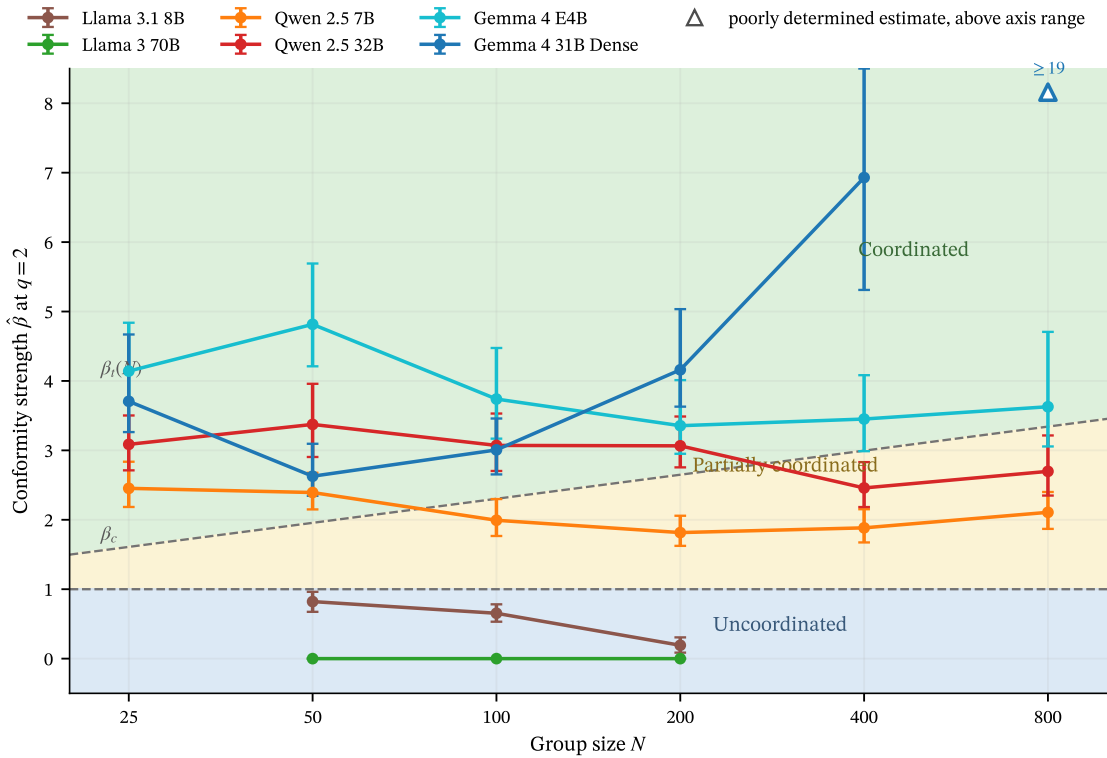


FIGURE 4.7: Fitted conformity strength at $q = 2$ against group size, in the phase-diagram form of the binary study (De Marzo, Castellano, et al., 2026). The label “Coordinated” denotes the theoretical regime where full coordination is expected, rather than an observed outcome. The lower dashed line is the critical value $\beta_c = 1$. The rising dashed curve is the fluctuation threshold $\beta_t(N)$. Error bars show 95 percent bootstrap intervals.

The graph reproduces the structure of the binary study. The two Llama models occupy the uncoordinated region, where Llama 3 70B collapses to zero, and Llama 3.1 8B lies below β_c at every measured group size and declines further as N grows.

The four coordinating models start inside the coordinated band, and their group-size trends repeat the findings of Figure 4.6 at $q = 2$: the conformity strength of Gemma 4 31B rises with N and moves deeper into the coordinated region. Gemma 4 E4B stays approximately flat, and the two Qwen models decline. Similar differences between models appear in the multi-option comparison, although the Qwen estimates do not decline consistently at $q = 10$. In the binary study, the conformity strength decreases with group size for most models, which produces a critical group size N_c beyond which full coordination becomes difficult to reach (De Marzo, Castellano, et al., 2026). For the models of this thesis the decline appears only in weakened form. The fluctuation threshold $\beta_t(N)$ increases with group size, whereas the fitted conformity strength of the Qwen models changes little. As a result, the Qwen values fall below the threshold. This occurs from $N = 100$ for Qwen 2.5 7B and from $N = 400$ for Qwen 2.5 32B. Their complete bootstrap intervals lie below the threshold at these group sizes. In the binary framework this means that a majority still forms but that the final approach to consensus slows down.

Figure 4.8 repeats this construction for the multi-option blocks at $q = 3, 10$ and 50 , with the critical value $\beta_c(q)$ as a horizontal line and the fluctuation threshold $\beta_t(N, q)$ rising with group size.

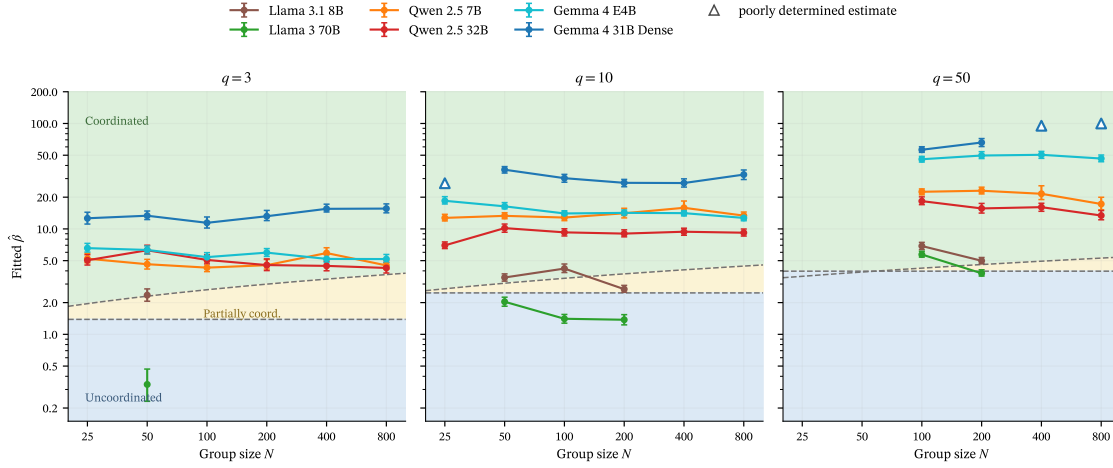


FIGURE 4.8: Fitted conformity strength against group size for $q = 3, 10$ and 50 . The horizontal dashed line is the critical value $\beta_c(q)$; the rising dashed curve is the multi-option fluctuation threshold $\beta_t(N, q)$.

Both axes are logarithmic.

The multi-option phase diagrams retain the broad differences between models seen

in the binary comparison. Gemma 4 31B generally increases with group size, Gemma 4 E4B varies less consistently, and the Llama estimates decline where several group sizes were measured. The Qwen estimates decline at $q = 3$ and $q = 50$, but not consistently at $q = 10$. In the displayed multi-option conditions, the Gemma and Qwen estimates remain above both benchmarks. Only the Llama models have estimates near or below them.

Qwen 2.5 7B does not reach consensus at $q = 50$ from $N = 100$ onwards, and Qwen 2.5 32B remains unfinished at $q = 50$ for $N = 400$ and 800 . Their fitted parameters remain above both benchmarks. The benchmark comparison therefore does not reliably distinguish completion within 100 sweeps under this protocol.

Each fitted $\hat{\beta}$ summarizes responses over a restricted set of states in which all q opinions remain represented. During the simulations, opinions disappear from the displayed list and the remaining support becomes distributed differently. The fluctuation threshold $\beta_t(N, q)$ also assumes a near-consensus state with a fixed set of available opinions. These differences limit the use of the initial- q fit to predict completion. The following section examines how the measured responses can still explain the observed outcomes.

4.3 Predicting Collective Outcomes from Measured Responses

The drift construction of Section 2.3 can turn measured response curves into predictions about where a population should come to rest. H3 asks whether such predictions hold in the simulations. Prediction serves here as the test of the experimental design. The question is whether the individually measured response, inserted into the framework of Chapter 2, accounts for them.

The Llama responses are applicable here, because the drift does not depend on a fitted response model. Instead, it uses the measured adoption probabilities directly. When the probability of choosing the leader is below the diagonal $P = s$, the leader is expected to lose support. When it is above the diagonal, the leader is expected to grow. For that reason, a crossing at which the drift changes from positive to negative can be seen as a predicted resting point. A leader share at which the population can come to rest without reaching consensus. Because the drift is built

from the measured probabilities, the comparison tests the design itself. If the states of the experimental design missed the behavior that decides collective outcomes, the predicted crossings would have no reason to match the simulations. The Llama models are well suited for this test precisely because they stall.

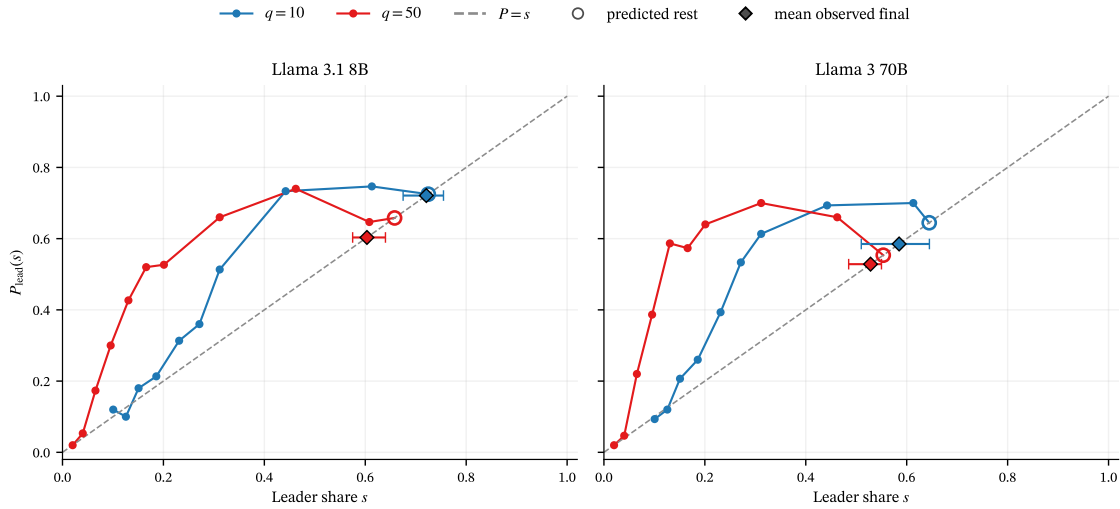


FIGURE 4.9: Measured leader-adoption probabilities of the two Llama models at a group size of $N = 200$, together with the zero-drift condition $P = s$ (dashed line). The measured curves are drawn up to their crossing with the diagonal. Open circles mark the predicted resting point. Diamonds mark the mean final leader share of the simulation runs.

For Llama 3.1 8B at $q = 10$, the predicted crossing at $s \approx 0.73$ almost exactly matches the mean final leader share of 0.72. This case is shown as an example. Across the other Llama conditions the crossing sometimes falls within the range of final leader shares and sometimes misses it by a few percentage points, and at the smaller group sizes the measured response of Llama 3.1 8B can stay above the diagonal up to the largest measured state, so that no resting point is predicted where one is observed. The comparison shows that the measured slice carries information about where these populations come to rest, but it is not a certain predictor.

In Figure 4.1, Qwen 2.5 7B does not reach full coordination at $q = 50$ and $N = 100$. The question is whether the controlled measurements predict this stall.

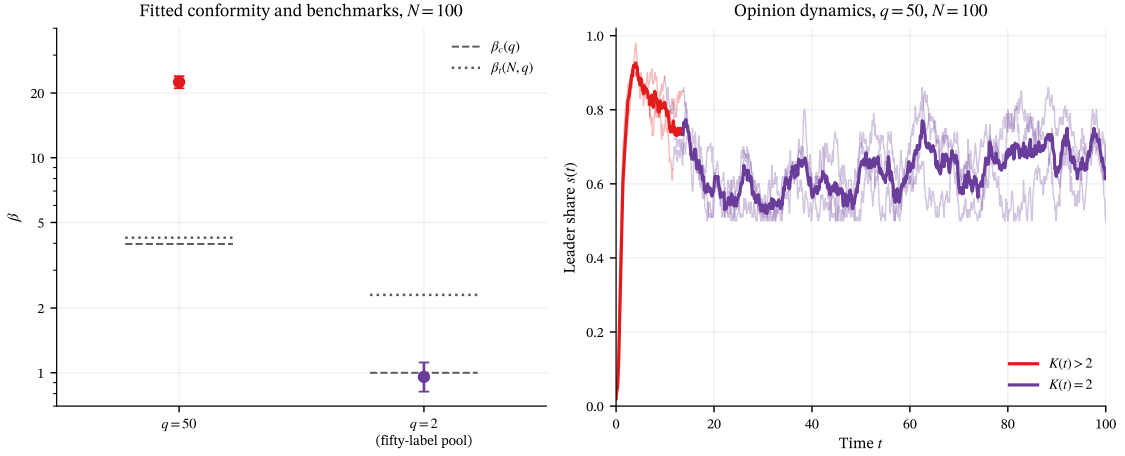


FIGURE 4.10: Qwen 2.5 7B at $N = 100$. Left: the pooled $q = 50$ fit and the supplementary binary fit using labels drawn from the fifty-label pool. Whiskers show 95 percent bootstrap intervals. Dashed and dotted marks show $\beta_c(q)$ and the fluctuation threshold $\beta_t(N, q)$. Right: three $q = 50$ trajectories and their mean, colored by the phase before and after two opinions remain.

The conformity measurement at $q = 50$ suggests that this condition should reach full coordination. The $q = 50$ fit at this group size is $\hat{\beta} \approx 22.5$, roughly five times both the critical value $\beta_c(50) \approx 3.97$ and the fluctuation threshold $\beta_t(100, 50) \approx 4.25$. The runs behave accordingly at first. They leave balance immediately within about the first time step. The stall itself occurs later, in the region where it contains extinct opinions and the initial- q measurements provide no direct observations. There only two of the fifty possible opinions remain occupied. Extinction reduces all three runs to two active opinions within 14 time steps, and the remainder of every trajectory is a binary endgame between the two survivors around balance, wandering between $s = 0.50$ and $s = 0.86$ for more than 85 time steps.

The opinion dynamics provide evidence for a transition between two regimes. The $q = 50$ populations first undergo a multi-option selection process, during which most opinions disappear. Their late stage contains only two active opinions and can be examined with the binary response framework. For a direct comparison, the binary response was measured once more at $N = 100$ with the two labels drawn from the fifty-label pool of the $q = 50$ condition rather than the fixed pair of the binary blocks, because this is what the surviving pair looks like in the run. The results follow a binary tanh response with $\hat{\beta}_{\text{bin}} \approx 0.96$. The interval overlaps the binary

critical value $\beta_c = 1$, and the estimate lies clearly below the fluctuation threshold $\beta_t(100) = \frac{1}{2} \ln 100 \approx 2.30$ linked to completion of consensus. The results support a binary interpretation of the endgame and account for the handover observed in the $q = 50$ trajectories, the populations leave balance as fifty-opinion systems and come to rest as binary ones whose measured response does not complete consensus.

Combined with the Llama results, this comparison shows that measured responses can help explain collective outcomes, although their usefulness depends on the states covered by the controlled design. Once opinions disappear and the leader share exceeds the measured range, the comparison needs to account for the remaining active opinions. The binary framework of De Marzo, Castellano, et al. (2026) may therefore also describe the late stage of multi-option coordination. Chapter 5 discusses this connection and its limitations.

4.4 Effective Conformity

The fitted value $\hat{\beta}$ summarizes the response across several controlled states. The effective parameter $\beta_{\text{eff}}(s)$ instead gives the value needed to reproduce the measured leader-adoption probability at one state. This allows us to examine whether the increase in fitted conformity with q also appears at small leads. Near balance, every opinion is still represented and the remaining support is nearly evenly distributed. These states provide a useful comparison across q and resemble the balanced starting conditions of the simulations, although later states can depart from this slice.

Figure 4.11 shows β_{eff} at the smallest controlled state above balance retained in the analysis for $N = 200$. For the displayed multi-option cases, this is $r = 0.03$. At $q = 50$, for example, the leader has 8 of the 199 displayed supporters, compared with 4 at $r = 0$.

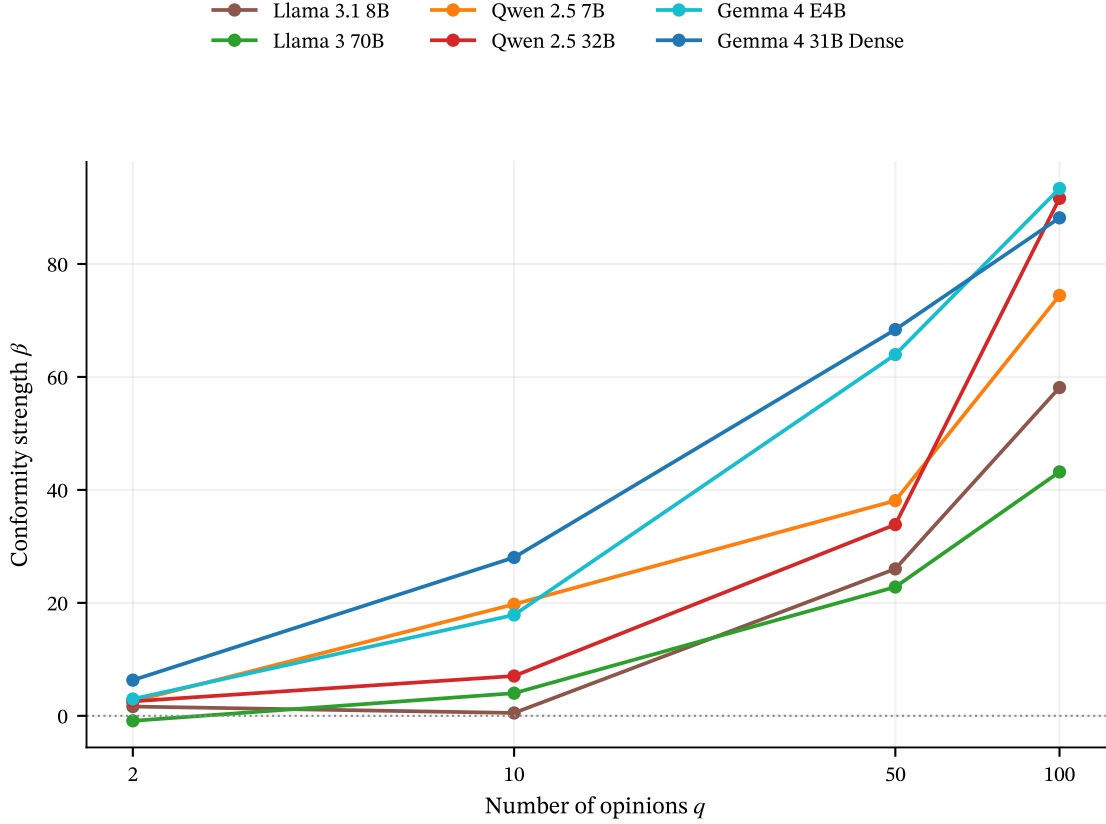


FIGURE 4.11: Effective conformity at the smallest retained measured lead against the number of opinions at $N = 200$. The displayed multi-option states use $r = 0.03$ and the binary states $r = 0.10$. Colors show the models.

Across the displayed multi-option cases, the effective parameter increases with q for all six models. For Qwen 2.5 32B, it rises from about 7 at $q = 10$ to 34 at $q = 50$ and 92 at $q = 100$; the corresponding Gemma 4 31B values are about 28, 68 and 88. For Gemma and Qwen, this is consistent with the increase already found in the pooled fits and with regard to these additional results, the trend also appears when the comparison is restricted to small leads.

The next comparison holds q fixed and follows the effective parameter as the leader's share grows. Figure 4.12 shows Qwen 2.5 32B at $q = 50$ and $N = 200$. None of the retained states has a leader-adoption frequency of zero or one, which allows the profile to be examined across the measured range with its Wald intervals.

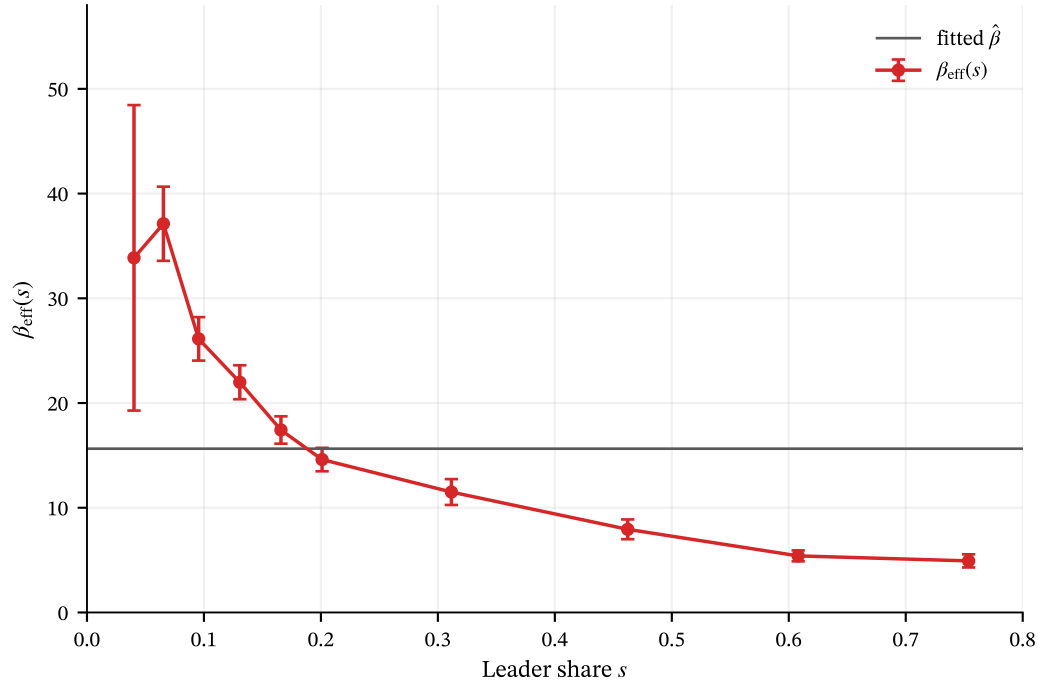


FIGURE 4.12: Effective conformity along the measured range for Qwen 2.5 32B at $q = 50$ and $N = 200$, with 95 percent Wald intervals on the logit scale. The solid line marks the pooled fit $\hat{\beta}$.

The effective parameter falls from about 34 at the smallest lead to about 5 at the largest measured share, a factor of seven, and the pooled fit of 15.6 runs through the middle of this decline. Reproducing the response therefore requires a larger effective β near balance than at more concentrated states. A single constant β cannot represent both parts of the curve: the pooled estimate understates the response near balance and overstates it at more concentrated states. A similar decline appears in other Qwen multi-option profiles. At $N = 200$, Qwen 2.5 7B falls from about 38 to 6 at $q = 50$ and from 74 to 12 at $q = 100$ across the states with adoption frequencies strictly between zero and one. Gemma responses often saturate too quickly to establish the same pattern across the measured range.

How the same profile looks in the binary case is shown in Figure 4.13, for the same model.

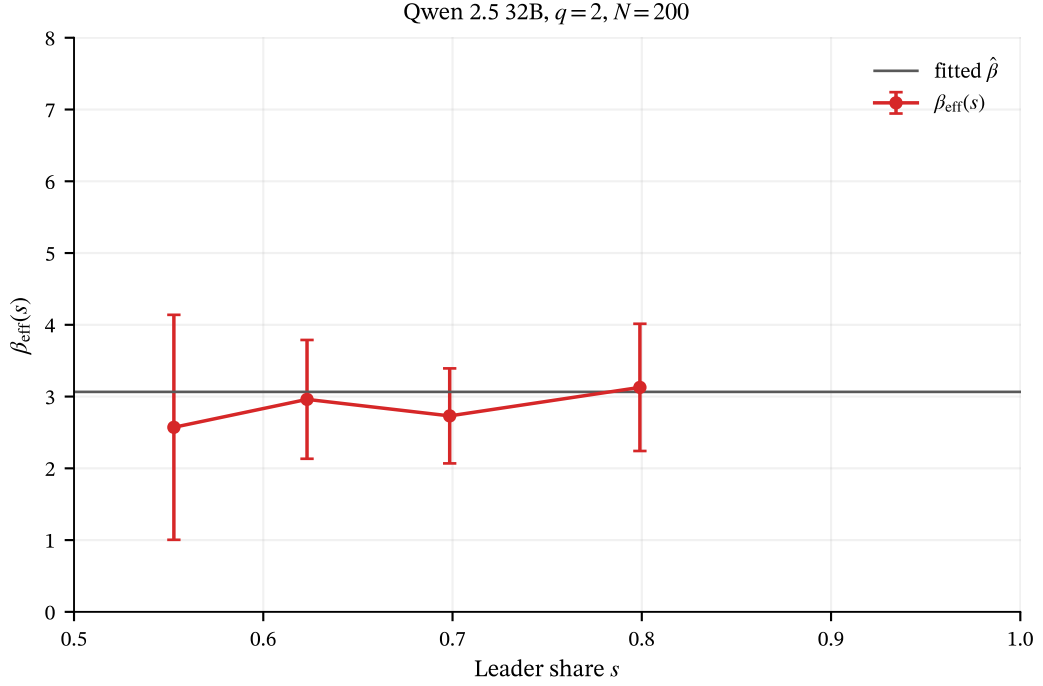


FIGURE 4.13: Effective conformity at $q = 2$ and $N = 200$ for Qwen 2.5 32B, with 95 percent Wald intervals on the logit scale. States in which the leader was chosen in every query are omitted.

At $q = 2$, the same model shows a different pattern. The effective parameter is compatible with a constant within its uncertainty across the retained states, and the pooled fit runs through the points instead of through a decline. One constant β is therefore an internally consistent description of this binary response. This provides a direct connection to the binary framework of De Marzo, Castellano, et al. (2026), while the decline at $q = 50$ shows where its constant-parameter description no longer holds.

Chapter 5

Discussion

5.1 Answers to the Research Questions

RQ1 asked whether coordination becomes more difficult as the number of available opinions increases. The Qwen results show sharp changes from rapid consensus to slow or unfinished runs as q increases. At $N = 200$, Qwen 2.5 7B completes all eight runs at $q = 3$, three of eight at $q = 10$, and none of three at $q = 50$ within 100 sweeps. Qwen 2.5 32B at the same group size completes all eight runs at both $q = 3$ and $q = 10$, compared with two of three at $q = 50$. Its three supplementary runs at $q = 100$ all finish, but with a median of about 51 sweeps compared with eight at $q = 10$. The difficulty also appears at smaller q in larger populations. At $N = 100$, Qwen 2.5 7B completes every run at $q = 10$ and Qwen 2.5 32B at $q = 50$. At $N = 400$ and 800, neither model finishes at these respective values of q , while both still finish at $q = 3$.

These changes resemble the loss of reliable completion at larger group sizes in the binary study (De Marzo, Castellano, et al., 2026), with the number of opinions providing a further dimension of the comparison. They suggest that the onset of difficulty depends on the model and group size, although the few tested values of q and the run counts do not locate the onset precisely. Both Gemma models reach consensus across the tested multi-option conditions up to $N = 800$ and $q = 50$, with no consistent ordering of consensus time by q . The Llama runs remain unfinished beyond $N = 25$ at every tested q , so their completion counts cannot distinguish an additional cost of more opinions there.

H1 is partly supported. Additional opinions make consensus slower or less likely in the Qwen conditions, while the Gemma results show no consistent cost over the

tested range. The Gemma and Qwen estimates increase with q and remain above $\beta_c(q)$, including in conditions that do not finish. These fits do not support the simple explanation of a fixed conformity strength falling below the rising critical value $\beta_c(q)$. Where the Qwen populations stall, a substantial lead has often already formed. Their difficulty concerns the completion stage, which the critical value does not describe.

RQ2 asked how majority-following changes with the number of available opinions. At a fixed displayed group size, the fitted Potts parameter increases strongly with q for the Gemma and Qwen models. The effective parameters at small leads show a similar increase across the displayed multi-option states. Thus, the expectation in H2 that the same β transfers across numbers of opinions is not supported.

The second part of H2 concerns constancy across opinion shares at a fixed q . Here the evidence varies by model and condition. For Qwen 2.5 32B at $q = 50$ and $N = 200$, $\beta_{\text{eff}}(s)$ falls from about 34 at the smallest retained lead to about 5 at the largest measured share, while the pooled fit is 15.6. The binary profile of the same model is instead compatible with a constant within its uncertainty. Gemma responses often saturate too quickly to resolve such changes precisely, while the Llama curves already depart from the assumed shape. H2 is therefore not supported as a general description.

RQ3 asks both whether the Potts softmax describes individual choices and whether measured responses predict collective behavior. The answer to its first part is model-dependent. The softmax approximates the Gemma and Qwen measurements reasonably well on the probability scale, but describes the Llama responses poorly. The state dependence identified under H2 also limits its interpretation as a single response rule.

On the second point, H3 receives partial support. The drift calculated from measured adoption probabilities identifies some observed states of partial coordination even when the Potts fit is poor. For Llama 3.1 8B at $q = 10$ and $N = 200$, the crossing near $s = 0.73$ agrees closely with the mean final share of 0.72. Other comparisons miss the observed shares or identify no crossing. Measured responses thus retain predictive information, but one controlled response curve does not account for the full path to consensus.

5.2 Limitations and Future Work

The Potts softmax is used here in the same spirit as the Curie-Weiss rule in the binary study (De Marzo, Castellano, et al., 2026), as a simple response rule with one conformity strength that can be fitted, compared with theoretical benchmarks and inserted into a drift construction. It is an approximation, and its usefulness does not depend on LLM populations being Potts systems. Several of its features are visible in the data. The Gemma and Qwen responses broadly follow the predicted sigmoidal shape with small RMSE on the probability scale. The rapid growth of a leading opinion is consistent with an ordering process, and the binary limit helps explain the late stage of the Qwen 2.5 7B runs at $q = 50$. What the fits do not provide is one parameter per model that transfers across numbers of opinions, across shares and across the states a run passes through.

The critical value $\beta_c(q)$ concerns equilibrium ordering, and $\beta_{\text{bal}}(q)$ concerns the growth of very small leads. The increase of the fluctuation threshold $\beta_t(N, q)$ with q gives a reason to expect more difficult completion at fixed conformity. This agrees with the direction of the Qwen difficulties, but the fitted values lie above the benchmark even in conditions that fail. The threshold comparison does not identify where completion becomes difficult.

The decline of $\beta_{\text{eff}}(s)$ means that reproducing different states within the Potts rule requires different parameter values. It occurs while all opinions remain represented, so extinction cannot explain this variation. It does not establish that an underlying tendency to follow others weakens as the leader grows.

One alternative rule compares the support of two opinions by its ratio rather than by its difference. In the Potts softmax, their relative odds depend only on the difference between their shares at a fixed conformity strength. Human judgments of numerical magnitude depend on ratios, as described by the Weber–Fechner law (Dehaene, 2003), and a model that reads a list of supporters may weigh support in the same way.

The Gemma 4 E4B measurements at $N = 200$ in Figure 4.5 motivate testing this alternative. Consider the smallest measured lead with three and with fifty opinions. With three opinions, the leader has 71 supporters and each rival 64. With fifty, the leader has 8 and each rival about 4. The difference is smaller in the second case, but the leader has about twice each rival’s support instead of only 11 percent

more. A Potts rule with fixed conformity strength predicts a larger advantage over an individual rival in the first case. A rule that responds increasingly to support ratios predicts the opposite ordering.

The measured choices show the second ordering. With three opinions, Gemma picks the leader 62 times out of 150 and each rival 44 times on average. With fifty opinions, it picks the leader 33 times and each rival about 2.4 times on average. Although the leader is chosen less often overall, its adoption frequency relative to the average rival rises from about 1.4 to 14. Gemma 4 31B and both Qwen models show the same direction of change. Other features of the display also change with the number of opinions, so this comparison does not isolate support ratios as the cause. Testing whether a ratio-based rule describes these responses more consistently would be a useful next step.

Opinion extinction adds another reason why one initial- q fit may not describe a complete run. The simulations lose many opinions early, while the controlled states keep all of them represented. Under the memoryless protocol, the current display is what matters. Tracking the number of active opinions $K(t)$ could connect the Potts description of the multi-option stage with the Curie-Weiss response of De Marzo, Castellano, et al. (2026) once two opinions remain. This approach would still require response measurements matched to the surviving opinions and their labels, including the appropriate conformity strength.

For Qwen 2.5 7B at $q = 50$ and $N = 100$, all runs reduce to two active opinions. The initial- q fit lies above its fluctuation threshold, while the binary measurement with labels from the fifty-label pool gives $\hat{\beta}_{\text{bin}} \approx 0.96$, with an interval spanning the binary critical value and lying below the binary fluctuation threshold. This supports the connection between the two descriptions in one case, although the binary measurement was collected after the stall was observed.

As Section 3.1 notes, randomization does not remove preferences for the labels themselves. With labels drawn from the fifty-label pool, the binary response of Qwen 2.5 7B is about half as strong as with the standard fixed pair. The $q = 50$ fit is still about ten times higher than the standard $q = 2$ fit. For this model, the comparison suggests that a common label pool would change the size of the rise with q without removing it.

Capability is understood here as knowledge and reasoning performance, for which benchmarks such as MMLU-Pro, an extended version of the Massive Multitask

Language Understanding benchmark, provide an external indicator (Wang et al., 2024). Published MMLU-Pro scores place Gemma 4 31B at 85.2 percent, while Gemma 4 E4B and Qwen 2.5 32B achieve similar scores of 69.4 and 69.0 percent, respectively (Google DeepMind, 2026; Qwen Team, 2024). Both Gemma models nevertheless coordinate more reliably in our experiments. Benchmark performance provides context for the association between capability and coordination reported by De Marzo, Castellano, et al. (2026), but does not by itself explain the differences in this small model selection.

Strong majority-following can help agents settle on a shared convention. In the controlled binary measurement at $N = 800$, Gemma 4 31B chooses the larger side of a 440-to-359 split in 149 of 150 queries. A modest lead can therefore receive an almost unanimous response, consistent with its rapid approach to consensus. With neutral labels, however, this shows how agreement can be amplified without evidence that the leading opinion is better. Such strong conformity brings back the concern raised in Chapter 1 that rapid agreement may leave little room to question an arbitrary or undesirable choice.

The measurements with semantic labels in Appendix B show a different side. With renewable energy and fossil fuels as opinions, Gemma 4 31B chooses renewable energy in most queries even when fossil fuels have 48 of 49 supporters. Other opinion sets produce different balances between preference and majority influence, as also found in the bias study (De Marzo, Bellina, et al., 2026). A response measured with neutral labels should therefore not be transferred unchanged to decisions about content.

The leader-even-rest slice does not cover all distributions at the same leader share. The structured-state test in Appendix A checks transfer to a few uneven distributions, but leaves the late stages with fewer active opinions untested. Future measurements could follow changing numbers of active opinions and competitor distributions to test the connection developed in Section 5.2.

Keeping every opinion represented also restricts the largest measured share (Section 3.3). With 200 agents, it is about 0.995 at $q = 2$ but only 0.754 at $q = 50$. The unmeasured region near consensus grows with q at fixed N .

Most controlled states contain 150 queries. These resolve large changes in adoption probability but give limited information about rare dissent. Selecting the leader

in every query does not establish that dissent is impossible. Bootstrap intervals quantify sampling uncertainty, not errors in the response rule.

The main multi-option conditions contain three to eight runs and stop after 100 sweeps. An unfinished run establishes only that its consensus time exceeds this horizon. These results are descriptive. More repetitions near the observed changeovers and longer runs would improve estimates of completion probabilities, while more queries near consensus would better resolve the remaining dissent.

The model selection was constrained by the available hardware. Most experiments ran on a single GPU with 80 GB of memory, and the largest checkpoint was used in quantized form. Local inference made repeated measurements at scale feasible, but the findings cover only six open-weight models from three families. Future work could extend the comparison through hosted APIs to models from providers such as OpenAI and Anthropic. This would test whether the observed response patterns and coordination limits also occur outside the present model selection.

Group size, supporter counts and prompt length change together, so their effects cannot be separated here. Explicit count summaries and a common label pool across conditions would provide comparison protocols.

The fixed- q Potts model allows opinions to lose all supporters and later reappear. In the LLM simulations, opinions without supporters disappear from the displayed list, although their labels remain valid replies. The main controlled measurements keep all q opinions represented, so they do not directly test responses when some opinions are absent. Using the fitted softmax at these states is an extrapolation. Additional measurements are needed to check its accuracy there.

The results concern homogeneous, memoryless populations under fixed inference settings. Comparisons with heterogeneous teams, agents with memory or systems exchanging arguments would test their transfer to other forms of interaction.

Chapter 6

Conclusion

This thesis examined how the binary framework of De Marzo, Castellano, et al. (2026) transfers to coordination among more than two opinions. Six open-weight models from three model families were studied through opinion dynamics simulations and controlled response measurements. The mean-field Potts softmax provided a candidate response rule for relating individual choices to collective behavior.

The simulations show rapid leader emergence and early opinion extinction. At $N = 100$, strong pluralities generally form within the first few sweeps, without a uniform slowdown as q increases. Populations starting with fifty opinions lose roughly half of them within about two sweeps. This early progress also occurs in conditions that later stall. Additional opinions therefore did not impose a uniform coordination cost: both Gemma models reached consensus across the tested multi-option conditions up to $N = 800$, while some Qwen conditions and most Llama conditions remained partially coordinated. The main difficulty often concerned the extinction of the last competing opinion.

Across conditions with reliable completion, consensus time generally increases with group size, while its dependence on the number of opinions is less consistent. The Qwen results give partial support to H1 through sharp increases in consensus time or prolonged disagreement as opinions are added, with difficulty appearing at smaller q in larger populations.

The controlled measurements identify a clear contrast between binary and multi-option responses. For Qwen 2.5 32B at $N = 200$, the binary effective parameter is compatible with a constant within its uncertainty, whereas the $q = 50$ profile declines strongly as the leader’s share grows. The Potts softmax describes the Gemma and Qwen measurements reasonably well on the probability scale, but their fitted values

also rise strongly with q . A constant parameter can thus describe selected binary conditions without providing a stable description across numbers and distributions of opinions.

The theoretical benchmarks help organize these comparisons, but do not reliably distinguish which populations complete consensus within the observation horizon. The Gemma and Qwen fits remain above the critical value $\beta_c(q)$, including Qwen conditions without consensus. Some Qwen conditions also fail to finish above the approximate fluctuation threshold. The measured response retains information that the pooled fit loses. For Llama 3.1 8B at $q = 10$ and $N = 200$, the measured zero-drift crossing agrees closely with the mean final leader share despite the poor Potts fit. Such agreement is not uniform, and transfer still depends on how closely the controlled states match those encountered in the simulations.

Opinion extinction connects the multi-option Potts framework with the binary framework of De Marzo, Castellano, et al. (2026). A population initially holding many opinions can reach a contest between two survivors. For Qwen 2.5 7B, the supplementary binary measurement using the fifty-label pool helps explain the disagreement observed late in the $q = 50$ runs at $N = 100$. The binary framework therefore remains a possible description of the late stage, provided that the measurement matches the current display, including its label pool. Future work could combine the two approaches by tracking the active opinions and measuring responses at their changing support distributions.

This thesis extends the binary framework to coordination among many opinions, with the mean-field Potts softmax as the multi-option counterpart of the Curie-Weiss response. The analogy carries over in form, the Gemma and Qwen responses follow the Potts shape, the benchmarks organize the fitted values, and the binary limit of the same rule describes the late stage of a multi-option run. What does not carry over is a single conformity strength per model, which rises with the number of opinions and changes along the response curve. The approach follows the perspective of collective machine behavior introduced in Chapter 1, which treats the interactions of machine agents as an object of empirical study, here through behavioral measurements combined with a mathematical model.

Code and Data Availability

The code used for the simulations, controlled response measurements, statistical analysis, and figures is available in the GitHub repository: <https://github.com/lorenzrck/Multi-Option-Coordination-of-LLM-Agents>.

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Appendix A

Design Validation

The controlled experiment measures responses along a leader-even-rest slice. This appendix checks whether a Potts parameter fitted on that slice also predicts choices when the remaining support is arranged differently. It tests the transfer of the fitted parameter.

Eight supplementary distributions were measured across $q = 5$ and $q = 10$, with $N = 50$. They include an even-rest control, a one-agent lead, graded support, a strong runner-up, and ties between two leading opinions. For example, the $q = 5$ control has counts $(20, 8, 7, 7, 7)$, while the strong-runner state has $(20, 17, 5, 5, 2)$. The leader’s share is unchanged, but its competition differs. The Potts softmax predicts lower leader adoption in the second state; the measurements check whether it predicts the size of this change correctly.

Each distribution received 150 valid responses per model. These supplementary responses were excluded from fitting. The comparison below focuses on the Gemma and Qwen models, since the Potts softmax already describes the Llama responses poorly on the main slice. Figure A.1 compares the predicted probabilities with the measured frequencies. Points close to the diagonal indicate agreement; points below it show that the softmax predicts leader adoption too often.

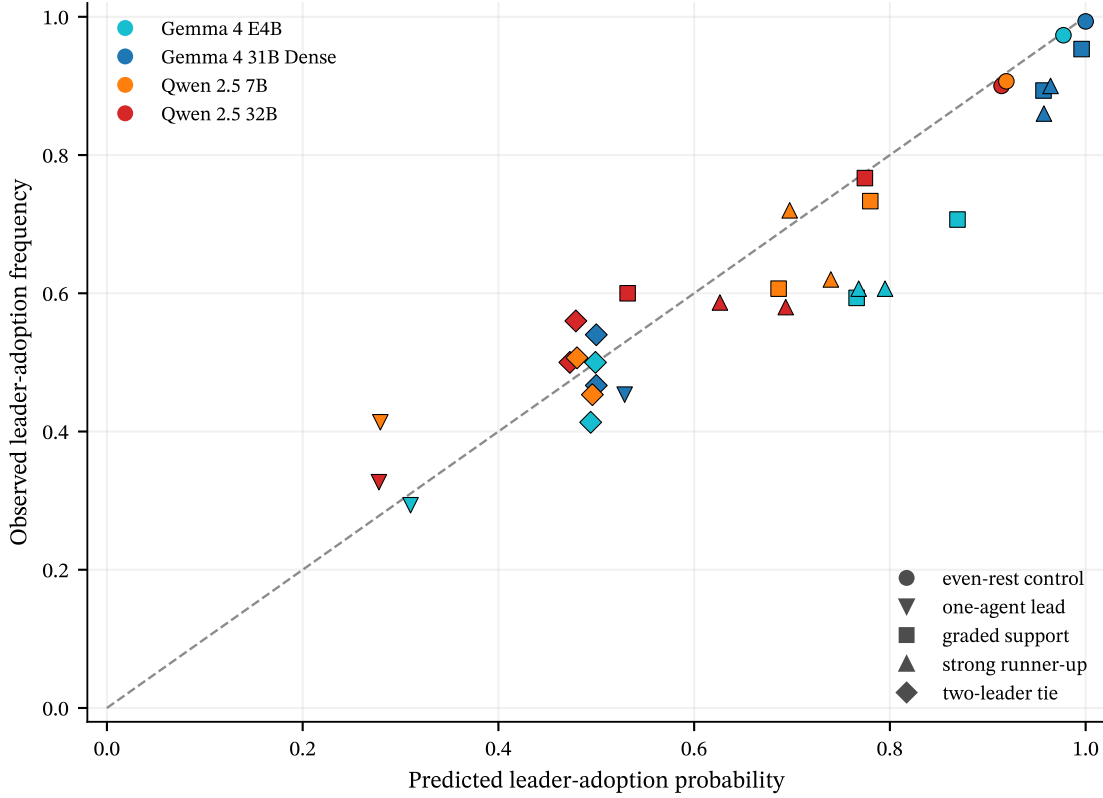


FIGURE A.1: Observed leader-adoption frequencies against predictions from the full-vector Potts softmax. Colors distinguish models and marker shapes the states.

The fitted parameter transfers approximately, but imperfectly, to the additional states. Many points lie near the diagonal which allows the conclusion, that the Potts rule captures much of the observed variation in leader adoption. The uneven distributions provide the main test of transfer beyond it.

The clearest departures are below the diagonal: the softmax often predicts leader adoption too frequently in states with a strong runner-up or graded support. This pattern is visible specifically for Gemma 4 E4B. Underprediction also occurs, particularly for the Qwen models in the one-agent-lead compositions and some tied states. As a result, the errors depend on both, the model and the arrangement of support and are not simply following one common direction. However, these results suggest that the simplified slice provides a useful basis for fitting the Potts softmax and approximately predicting multi-option choices beyond the fitting states for the models and conditions tested.

Appendix B

Semantic labels: preference and conformity

The main experiments use neutral labels. As Chapter 4 shows, even these labels are not free of preferences. Semantic labels introduce a further question, which is how an agent responds when the majority supports an opinion it tends to avoid. This appendix examines that question using the controlled design with semantic labels, following the bias study of De Marzo, Bellina, et al. (2026). Five of the opinion sets are presented here as additional response measurements. Energy sources ($q = 2$), speech ($q = 3$), AI governance ($q = 4$), political ideology ($q = 6$), and policy priorities ($q = 10$).

All five sets were measured for the six models with 50 agents. The prompt format and inference settings were those of Chapter 3, with semantic labels replacing the neutral labels. The measurements use a seven-point grid, $r \in \{0, 0.1, 0.25, 0.4, 0.6, 0.8, 1\}$. At each position, support outside the designated leader was divided as evenly as possible.

The semantic identity of each opinion was retained, while the leader role rotated among the opinions. Each opinion therefore occupied that role approximately equally often, allowing its response to be measured at different levels of support. Across the five sets and six models, 41,580 queries were collected, of which 40,998 were valid (98.6 percent). The probabilities below are calculated from valid replies.

B.1 Binary comparison: energy sources

The energy responses are fitted with the biased binary rule of De Marzo, Bellina, et al. (2026):

$$P(m) = \frac{1}{2}[1 + \tanh(\beta(m + h))].$$

Here, $P(m)$ is the probability of choosing renewable energy, and m is its displayed support share minus the share supporting fossil fuels. The parameter β controls the response to support, while a positive h shifts the curve toward renewable energy. With $h = 0$, the rule reduces to the neutral binary response in Chapter 2. The parameters were estimated jointly by maximum likelihood from valid replies. The fitting code uses the additive-field form $P_\alpha \propto \exp(2\beta m_\alpha + h_\alpha)$, where h_1 and h_2 are the fields for renewable energy and fossil fuels. For $\beta > 0$, the binary bias is $h = (h_1 - h_2)/(2\beta)$, giving the response written above. The neutral comparison uses the same seven controlled states and displayed group size.

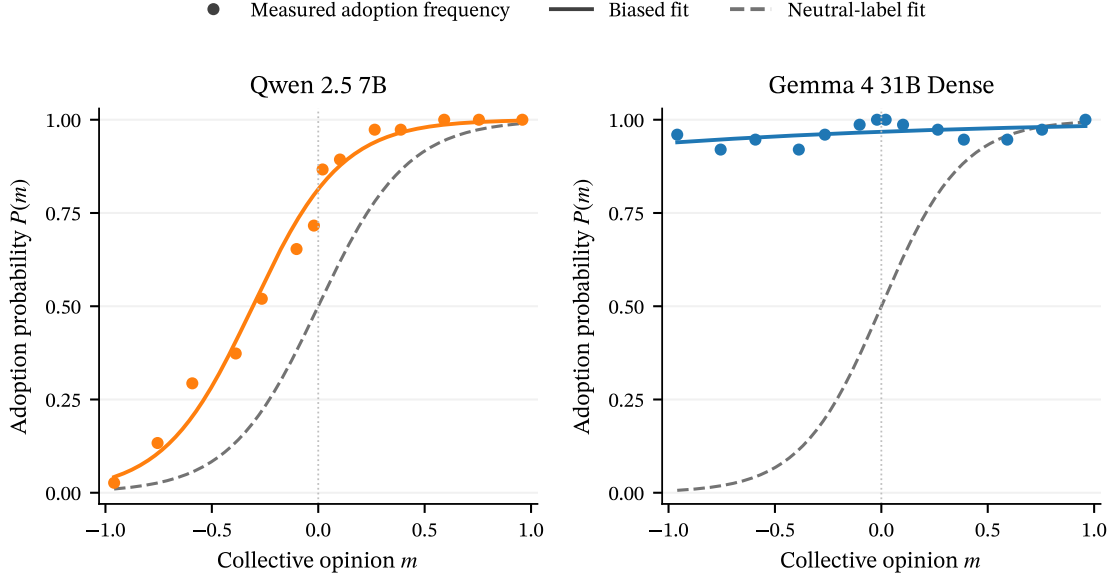


FIGURE B.1: Binary energy comparison at $N = 50$. Points show the measured probability of choosing renewable energy over fossil fuels; solid lines show the biased fits. Negative m means that fossil fuels have more supporters. Dashed lines show the corresponding neutral-label fits on the same grid.

The energy pair provides a direct connection to the bias study (De Marzo, Bellina, et al., 2026). Figure B.1 shows two contrasting responses. Qwen 2.5 7B favors renewable energy at similar levels of support, but a sufficiently large fossil-fuel majority changes its choice. When fossil fuels have 48 of the 49 supporters, it chooses renewable energy in only 2 of 75 queries (2.7 percent). Gemma 4 31B chooses renewable energy in 72 of 75 queries (96 percent) at the same state. For this model, even the largest displayed fossil-fuel majority rarely outweighs the preference for renewable energy.

For Qwen 2.5 32B, the bias study reports $\beta = 2.48$ for the energy pair, close to the estimate of 2.44 here (De Marzo, Bellina, et al., 2026; De Marzo, 2026). Both fits favor renewable energy strongly enough that the binary model predicts no persistent fossil-fuel majority. The measurements use the same checkpoint, opinion pair, prompt format, group size and temperature, although the state grid, nucleus sampling settings and fitting procedure differ. The agreement with the binary study provides an additional validation check for the response measurements, similar to the comparisons in Chapter 3.

B.2 Majority influence across multiple semantic opinions

Figure B.2 and Figure B.3 show how the adoption of each opinion changes in the multi-option case as its support increases. Each curve follows one opinion in the designated leader role while the remaining support is divided as evenly as possible among the other opinions. The lines connect measured adoption frequencies; they do not show fitted softmax responses.

The ideology curves show clear differences between opinions. Qwen 2.5 32B chooses socialism in only 7 of 40 replies when it has 44 of the 49 displayed supporters, but chooses nationalism in all 40 replies under the corresponding distribution. Gemma 4 E4B follows every ideology at the largest lead, whereas Llama 3.1 8B rarely follows several of them. Llama 3 70B remains concentrated on liberalism, whose adoption also falls at the strongest support levels similar to the behavior in the neutral label runs. A large majority does not seem to have the same effect for every opinion within a set.

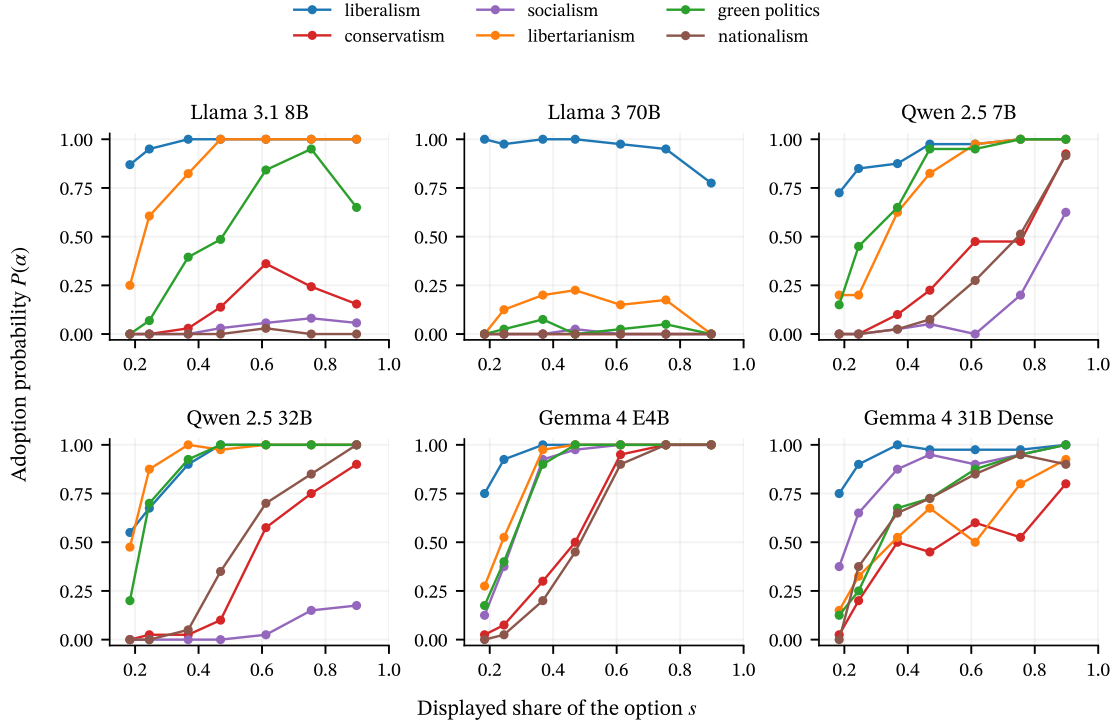


FIGURE B.2: Responses in the political-ideology set ($q = 6$, 49 displayed agents). Each curve shows the frequency of choosing an opinion as its displayed share s increases in the designated leader role, with the remaining support divided as evenly as possible.

The comparison with energy makes the dependence on content particularly clear for Gemma 4 31B. It chooses renewable energy in 96 percent of replies even against a 48-to-1 fossil-fuel majority. Yet it chooses nationalism and conservatism in 90 and 80 percent of replies, respectively, when those opinions have their largest ideology majorities. Its resistance to a majority in the energy measurement does not extend uniformly to other topics.

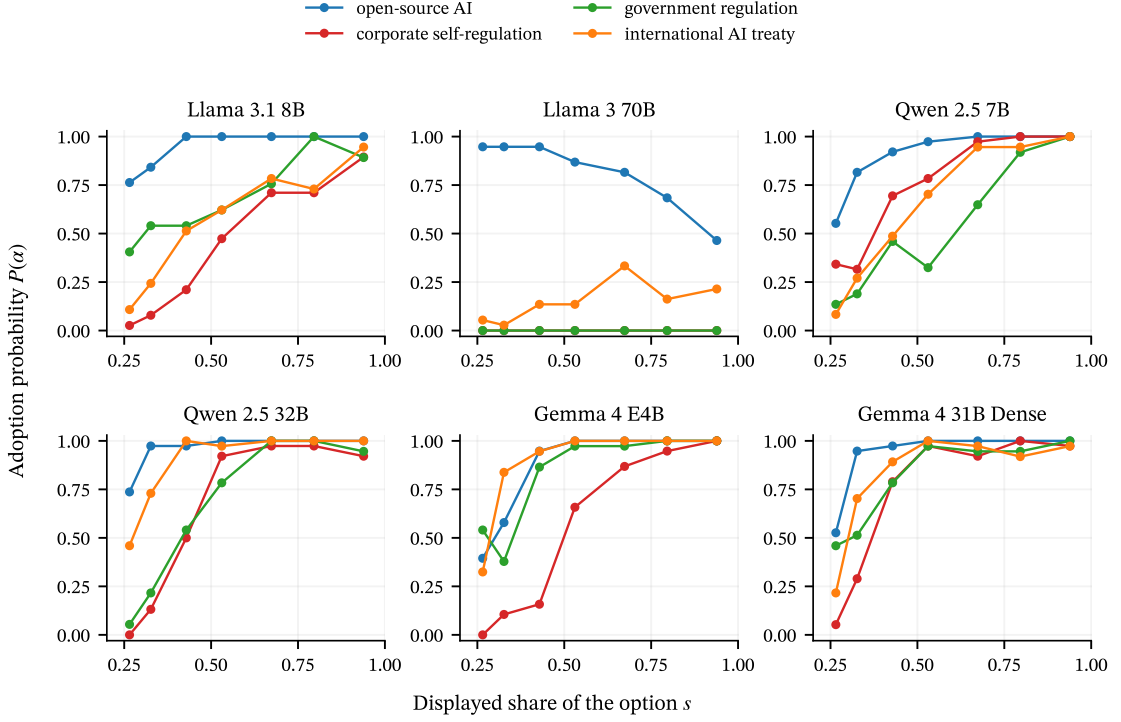


FIGURE B.3: Choice responses in the AI-governance set.

Strong support often outweighs the preferences visible near balance even more here. In the AI-governance set, both Gemma and both Qwen models choose every opinion in at least 92 percent of valid replies at its largest displayed share. This includes corporate self-regulation, which three of these four models rarely choose near balance. The pattern extends beyond governance. Across the four multi-option opinion sets, these models choose the leading opinion in at least 90 percent of valid replies in 89 of the 92 opinion-model combinations at the largest measured lead.

The curves retain some patterns seen with neutral labels in Chapter 4. Both Gemma and both Qwen models often follow an opinion strongly when its support is large, while Llama 3 70B again shows declining adoption at the largest shares. Semantic labels also produce differences between opinions within the same opinion set. These observations show why a response measured with neutral labels should not be transferred unchanged to opinions with content.

Each opinion has its own preference term in the multi-option fit:

$$P(\alpha \mid m) = \frac{\exp(2\beta m_\alpha + h_\alpha)}{\sum_{\gamma=1}^q \exp(2\beta m_\gamma + h_\gamma)}.$$

Here, h_α describes preference for opinion α , while one β controls the response to support. Parameters are fitted by maximum likelihood for each model and opinion set, with one preference term fixed at zero because only differences between the terms matter.

An exploratory fit of the biased Potts response shows that its accuracy depends on the model and the opinions being compared. The fitted curves can therefore shift along the support axis, but with equally divided remaining support they retain the same logistic shape and steepness. Across the five opinion sets, query-weighted RMSE ranges from about 0.01 to 0.11 for Gemma 4 E4B and the two Qwen models. Larger errors occur for Gemma 4 31B in political ideology and Llama 3.1 8B in policy priorities, reaching 0.17 and 0.19. For Llama 3.1 8B, the ideology fit predicts that conservatism is chosen in 54 percent of replies at its largest lead, compared with the observed 15 percent. Llama 3 70B’s declining responses also depart from the rule. As in Chapter 4, the fitted response provides an approximation whose adequacy varies across conditions.

The semantic measurements share the restrictions of the neutral design. Each number of opinions here corresponds to a different topic, so these measurements cannot answer whether additional opinions slow coordination. Predicting those dynamics would also optionally include responses at distributions beyond the measured slice.

In spite of that, the measurements show that strong support often outweighs the preferences visible near balance. Both Gemma and both Qwen models follow most opinions at their largest displayed shares in the multi-option measurements, although the exceptions depend on the model and the particular opinion. Within the same group of political ideologies, for example, Qwen 2.5 32B follows nationalism readily while rarely choosing socialism. This extends the competition between preference and conformity described by De Marzo, Bellina, et al. (2026) to choices among several opinions.

These initial multi-option measurements support the concern raised by De Marzo, Bellina, et al. (2026) that presented opinions of others in a group can redirect individual choices toward opinions that are rarely selected otherwise. The tests cover one group size and do not establish whether such choices persist in interacting dynamic groups. Even within this limited design, it became clear, that the influence of displayed majorities extends beyond binary decisions. For AI safety, these findings

strengthen the case for testing responses to fabricated majorities alongside individual alignment evaluations, before models are deployed in interacting systems.

Appendix C

Abliterated model comparison

This appendix compares coordination behavior and opinion preferences in the unaltered model, Qwen 2.5 7B Instruct, and the abliterated model, huihui-ai’s abliterated-v2 release (huihui-ai, n.d.). Abliteration targets refusal-related representations in the model (Arditi et al., 2024) and removes them.

The comparison follows Chapter 3’s protocol, with three initially balanced runs per condition and 100 time steps.

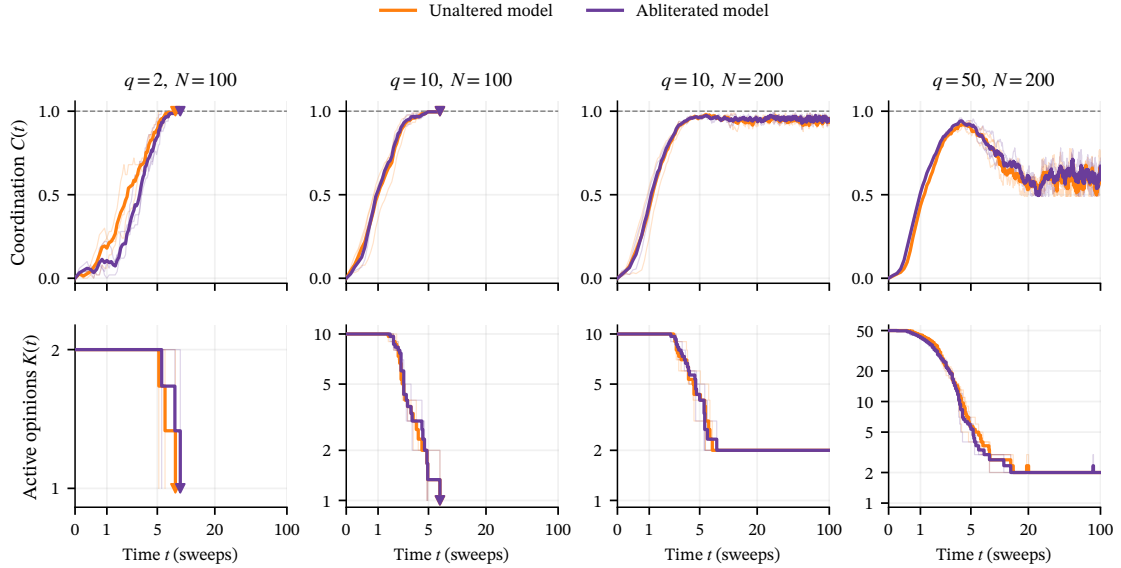


FIGURE C.1: Neutral coordination dynamics. Thick curves average three runs; thin curves show individual runs. Completed runs are held at consensus. Time uses $\log(1+t)$ and $K(t)$ a logarithmic scale.

The completion pattern is the same for both models (Figure C.1). All runs in the tested $N = 100$ conditions finish, while none at $N = 200$ does. At $q = 2, N = 100$,

mean consensus time rises from 6.37 to 7.40 sweeps in the abliterated model; at $q = 10, N = 100$, both means are 5.42. The stalled populations first reach two active opinions within 15 sweeps. At $q = 10, N = 200$, they remain close to consensus, whereas at $q = 50$ the early strong lead falls back to a less coordinated state.

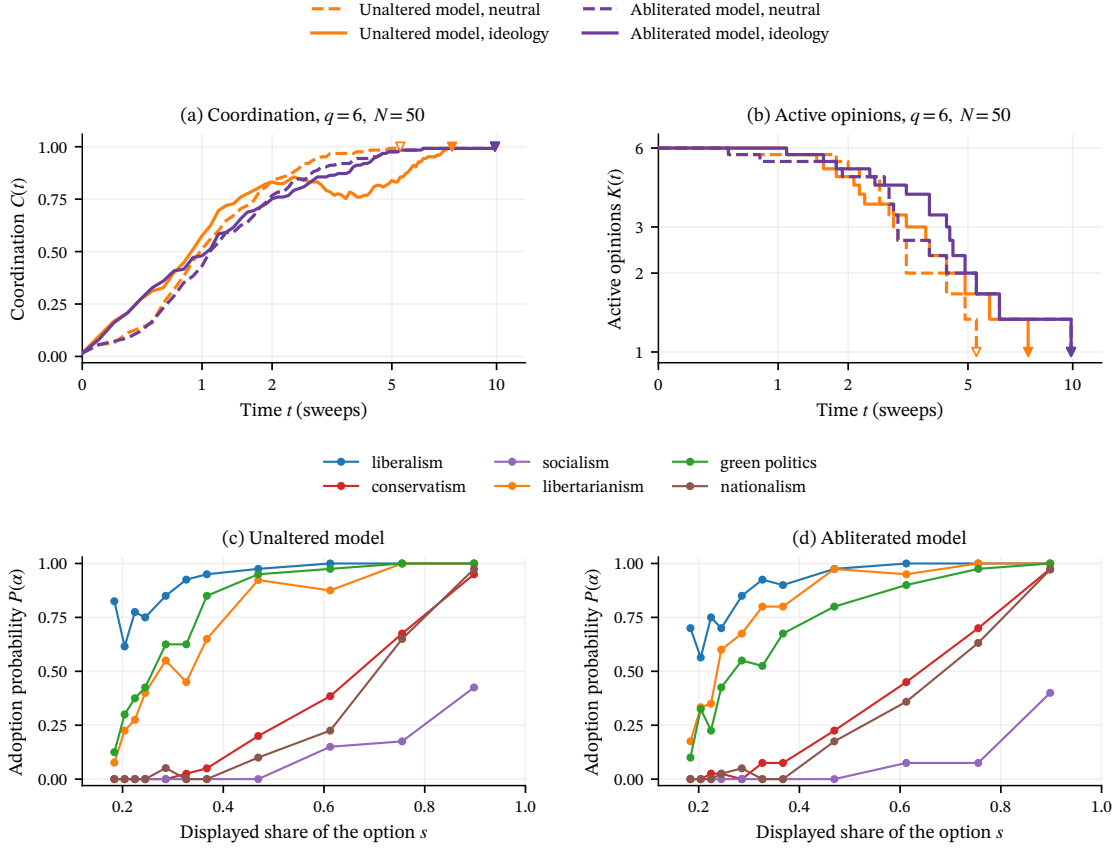


FIGURE C.2: Matched semantic comparison at $q = 6, N = 50$. (a–b) Mean dynamics over three runs, with neutral controls dashed and ideology runs solid. (c–d) Adoption frequencies for each ideology in the designated leader role.

Every run at $q = 6, N = 50$ reaches consensus with both neutral and ideology labels (Figure C.2 a–b). Although the ideology curves rise earlier, mean consensus time increases from 4.18 sweeps with neutral labels to 5.82 with ideologies for the unaltered model. The abliterated model's means are almost the same, 6.45 and 6.53 sweeps. Semantic labels therefore prolong coordination mainly in the unaltered model runs. The final opinion is liberalism in two runs with the unaltered model

and green politics in the third. All three runs with the abliterated model end on liberalism.

Both models favor liberalism at the smallest displayed lead, while socialism remains rarely selected across much of the response curve. Even with 44 of 49 supporters, socialism is adopted in only 17 of 40 replies from the unaltered model and 16 of 40 from the abliterated model. Some adoption frequencies differ between the models. At 16 of 49 supporters, libertarianism is selected in 18 of 40 replies from the unaltered model and 32 of 40 from the abliterated model. At 18 of 49 supporters, adoption of green politics falls from 34 of 40 of the unaltered model to 27 of 40 of the abliterated one. These differences suggest shifts in responses to particular ideologies, although individual points are based on only 40 replies.

Strong ideological preferences nevertheless remain in both models. The abliterated model also reaches consensus in the same tested conditions as the unaltered model and stalls in the same conditions. In this comparison, targeting refusal-related representations leaves the broad coordination pattern intact and does not eliminate the strong dependence of individual choices on opinion content.